

Particulate Matter

Week in Review

Week ending 12 September 2026 | 6 September 2026 to 12 September 2026 | 126 records pooled from 7 daily issues

126

RECORDS POOLED
(126 UNIQUE DOIs)

40%

MEASUREMENT SIDE
(51 RECORDS)

10

SUBTOPICS
ACTIVE

47

POOLED EFFECT
ESTIMATES

20

ABSTRACT-FREE
(16% OF WEEK)

The week in one paragraph

Sensing led a week for the first time since mid-August, the quantitative epidemiology came back hard after last week's floor — and one in six records this week has no abstract in existence anywhere. The pipeline's best-performing leg and its worst failure mode are now the same leg.

Scale. 126 records over seven issues on daily totals of 15, 20, 18, 26, 16, 10 and 21, a far flatter week than the 5.4× range of the last one. Against recent rollups: 126 after 150, 109, 112 and 134 — squarely mid-range, and the smoothness matters more than the level, because week-over-week cluster movements below are not being read through a single 38-record deposit day.

Sensing, forecasting & instrumentation at 31 is the largest cluster, and that is the week's structural news. For three consecutive weeks *Exposure assessment & modelling* had led; it has now halved, 47 → 20, the sharpest single-cluster contraction this watch has recorded, while Sensing fell only 34 → 31. The measurement side as a whole went 54% → 40%, so the two clusters did not move together: modelling-side exposure work collapsed and instrumentation held.

Estimate density recovered to one per 2.7 records. 47 pooled estimates against 126 records, from a four-week series of 2.7, 2.9, 4.4, 2.7. Last week's rollup flagged 4.4 as the weakest on record and declined to say whether it was a real thinning or a measurement-heavy artefact. One week later it is back to the series norm, which settles it: **it was the artefact**. The week's quantitative spine is cardiometabolic — CHARLS coexposure, the Zhongshan AMI case-crossover, the Shijiazhuang AECOPD series, the Hangzhou aortic-dissection component analysis, the Korean 8.25-million asthma cohort and the Kadoorie fracture cohort.

The abstract-free problem is now large enough to be a finding about the corpus. 20 of 126 records (16%) shipped as METADATA ONLY, up from 11% last week, and 10 of those 20 are in *Sensing* — the pooled heatmap's densest cell is *Sensing × Metadata only*. Crossref-by-ISSN is the only leg that reaches *J. Aerosol Sci., Atmos. Res., AMT* and *ACP*, and it indexes a DOI days before the publisher deposits the abstract. So the leg that makes the instrumentation literature visible at all is the same leg that delivers it unreadable. **An abstract-retry pass over the standing list is the highest-value pipeline work outstanding** — it would convert roughly a sixth of the week from title-only to summarised.

Subtopic trend and inflection

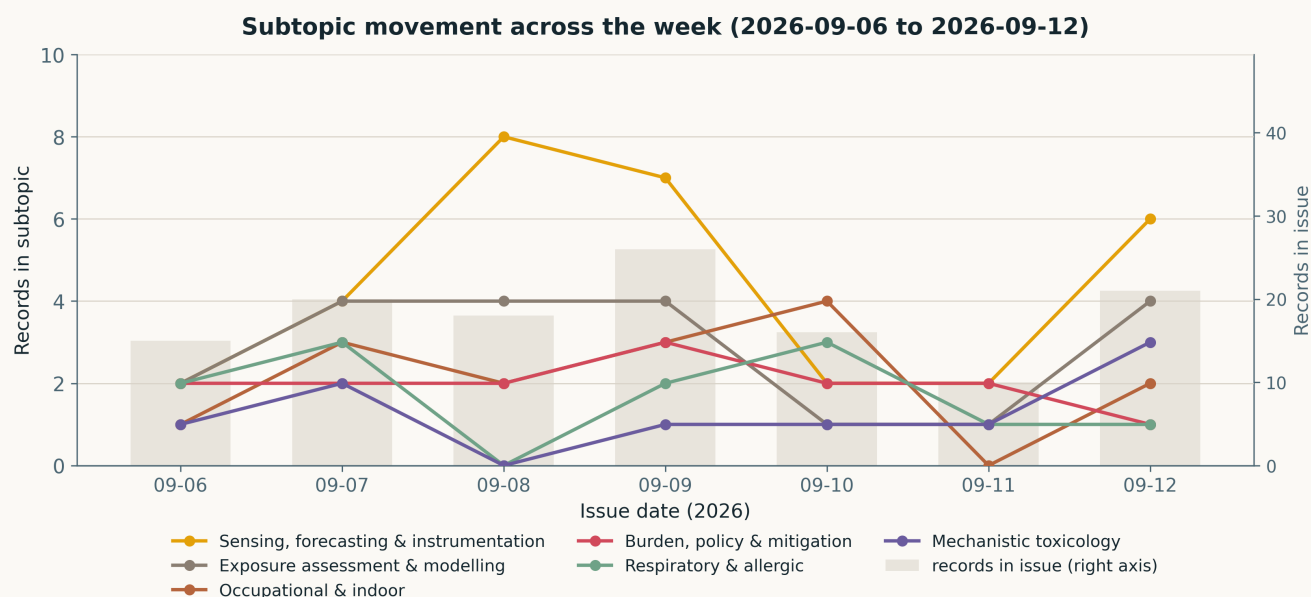


Fig. Per-issue subtopic counts across the week from `state/metrics.csv`, seven issues. Daily totals were 15, 20, 18, 26, 16, 10 and 21 — a $2.6\times$ range against $5.4\times$ last week, so the per-cluster lines are readable as movement rather than as denominator noise. Two of the seven days are weekends (6 and 12 September) and both are PubMed-suppressed: 12 September admitted **26 records registry-wide** against 1,135 on 8 September. Those two days were carried almost entirely by Europe PMC and Crossref-by-ISSN, which is why the *Sensing* line does not dip at the weekend the way the health clusters do.

Expanding. *Burden, policy & mitigation* 7 → 14 doubles off last week's recorded floor and is the week's largest proportional gain. *Respiratory & allergic* 5 → 12 likewise more than doubles from a floor. *Neuro / mental health* 4 → 7 and *Cardiovascular & metabolic* 7 → 8 both continue upward. Read together, the health side recovered as a bloc from a single unusually thin week — which is the strongest evidence available that last week's health-side floor was a sampling artefact and not a turn in the field.

Contracting. *Exposure assessment & modelling* 47 → 20 is a halving from a cluster record, and no interpretation of it should outrun that fact: a record high is the most likely thing to be followed by a fall. *Occupational & indoor* 19 → 15 eases off last week's surge but stays well above the pre-surge regular-week maximum of 9 — **the indoor turn the 5 September rollop declined to call a trend has now held for a second week**, which was the stated test. *Mechanistic toxicology* 11 → 9 and *Sensing* 34 → 31 are both mild.

The inflection worth naming. The four-week measurement-side share now reads 41, 47, 54, 40% — it oscillates, it does not trend, and the 29 August rollop's warning against building a claim on it stands. What has changed is *within* the measurement side: *Exposure* and *Sensing* moved in opposite directions relative to each other for the first time in four weeks. One week is not a pattern. The concrete thing to watch next week is whether *Exposure* returns to the mid-20s, which would confirm this week as the correction of a record rather than a shift.

Study architecture

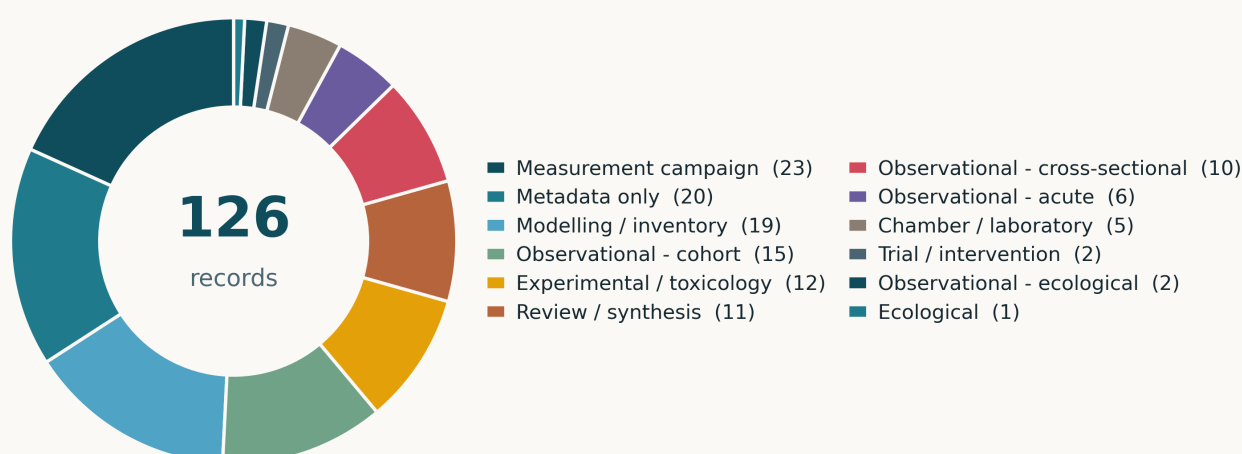


Fig. Study architecture over the pooled week. *Measurement campaign* (23) retakes the lead from *Modelling / inventory* (19), reversing the first-ever modelling-ahead-of-measurement week recorded in the last rollop. *Metadata only* at 20 is the second-largest architecture in the week, which is not an architecture at all but an absence of one. Primary observational epidemiology (cohort 15, cross-sectional 10, acute 6) totals 31, ahead of modelling for the first time in three weeks. **Two** interventional records appear in the 126 — a Dutch school air-cleaner RCT and a smoke-free-home trial in public housing — against one last week.

Pooled analytics

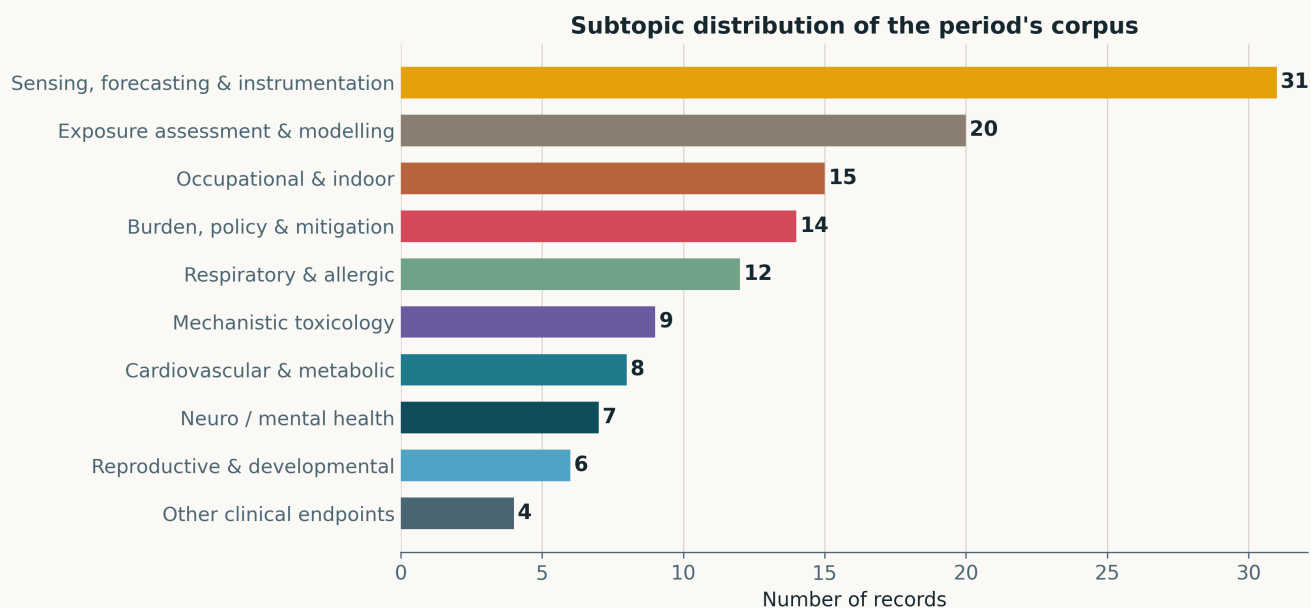


Fig. Pooled subtopic distribution, 126 records. All ten clusters were active, for the seventh consecutive week. The two measurement clusters (31 + 20 = 51) no longer outweigh the eight health and policy clusters combined (75) — last week they did, 81 to 69.

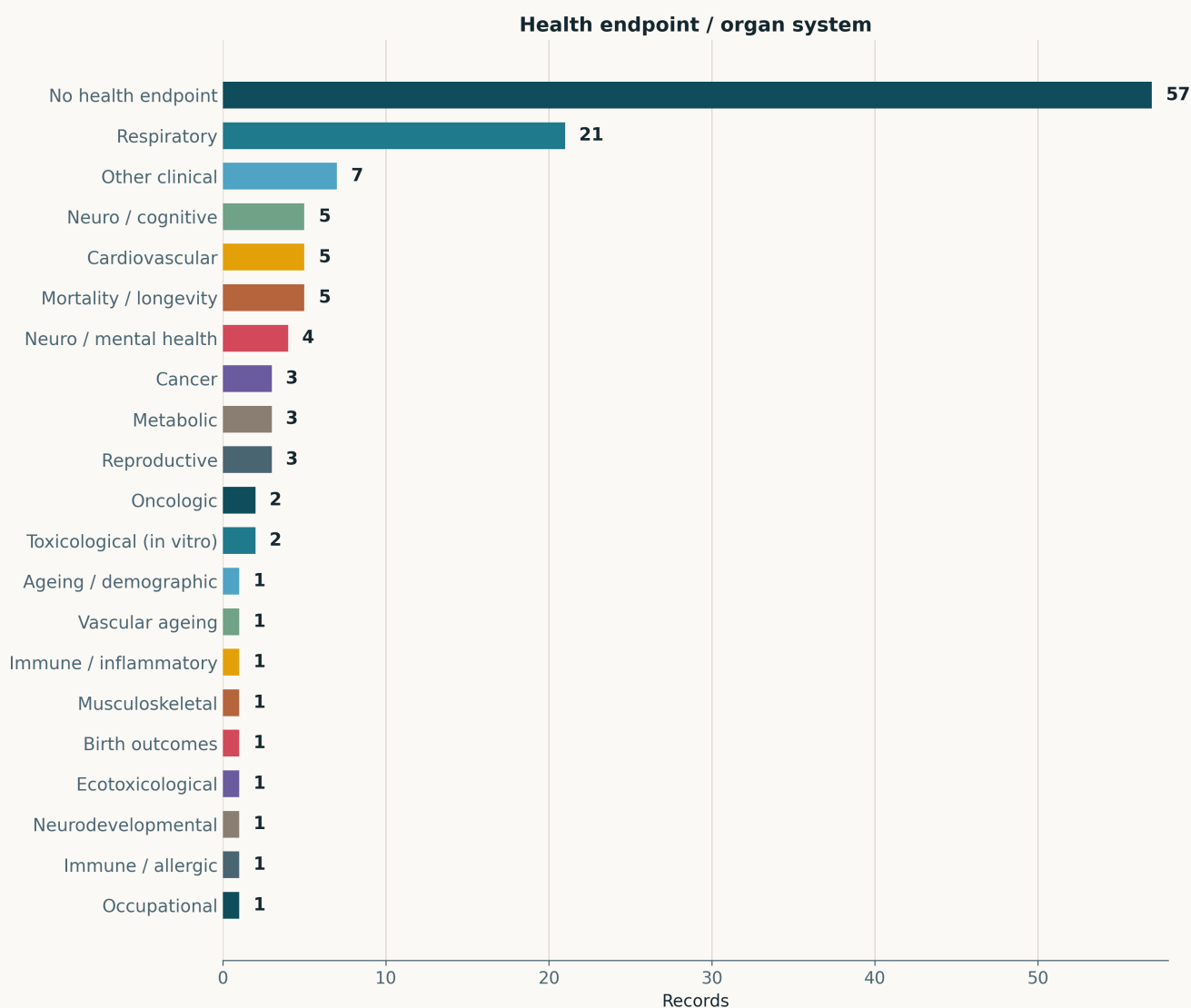


Fig. Health endpoint by organ system. 55 of 126 records carry no human endpoint — monitoring (27), instrument (9), emissions (6), aerosol chemistry (5) and two explicit none-labels — a 44% share, close to last week's 38%. Among endpoints that exist, *Respiratory* at 21 is far ahead of anything else and is more than

the next three combined (mortality 5, cardiovascular 5, other clinical 5). Note that *Cancer* and *Oncologic* remain split in ENDPOINT_CANON, so oncological work is understated by this panel; logged 3 September, still unfixed.

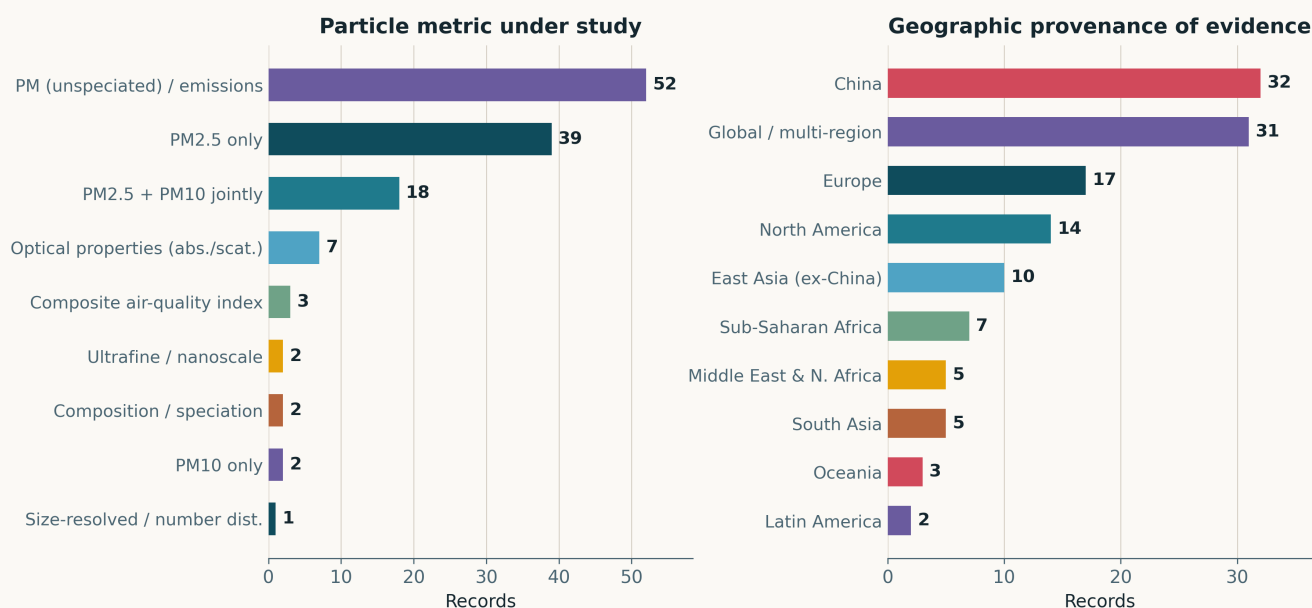
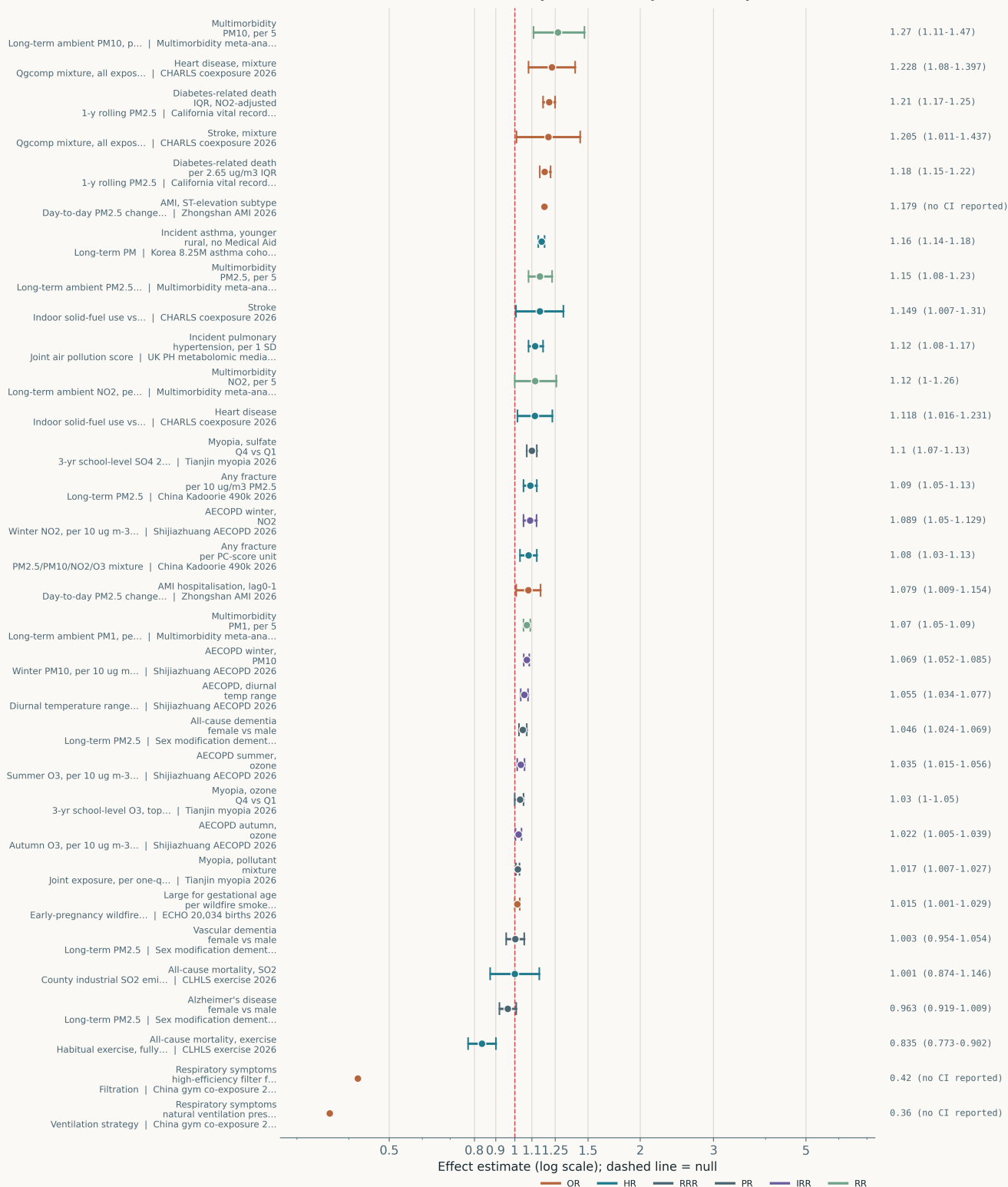


Fig. Left — particle metric. Unspeciated PM and emission factors (52) lead PM_{2.5}-only (39), with PM_{2.5}+PM₁₀ jointly at 18. **Ultrafine and size-resolved number together total 3 of 126**, against seven records whose metric is an optical property — the field is measuring what instruments report, not what the source signal lives in. Right — geographic provenance. **China supplies 32 of 126 (25%)**, with the *Global / multi-region* fallback at 31, Europe 17, North America 14 and East Asia (ex-China) 10. Sub-Saharan Africa at 7 is the strongest week this archive has recorded for the region. The fallback bar is inflated by laboratory and computational records that have no site by construction, and by daily-issue geography labels that were never normalised — that defect is carried and is logged again below.

Quantitative associations reported in the period's corpus (1 of 2)



Quantitative associations reported in the period's corpus (2 of 2)

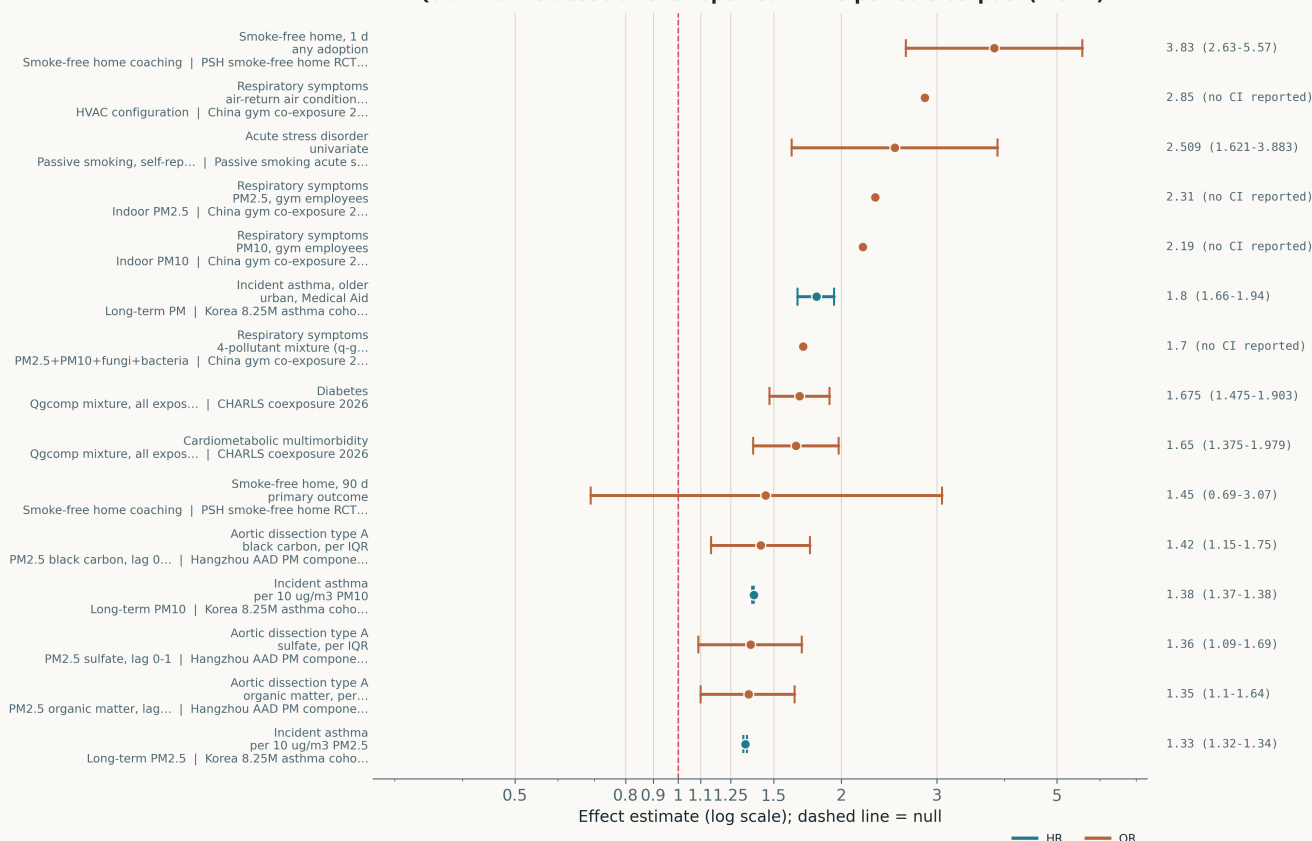


Fig. All 47 pooled effect estimates on a common logarithmic axis, split across two panels. **These are not mutually comparable and are not meta-analysed:** contrasts include per-IQR, per-10 $\mu\text{g m}^{-3}$, per-quartile, per-SD, per-smoke-day and mixture-index units, on OR, HR, RR, IRR, PR and RRR. **Heterogeneity is structural, not statistical**, so no I^2 is reported; pooling these would be a category error. Three things are readable anyway. First, **four estimates strictly cross the null** — the CLHLS SO_2 mortality hazard, the Alzheimer's and vascular-dementia sex-modification ratios, and the smoke-free-home 90-day primary outcome — with two more whose lower bound sits exactly at 1.00 (ozone–myopia, NO_2 –multimorbidity), against two of 34 last week. A further **seven estimates carry no interval at all** because the abstract reported none; they are drawn as bare points and should not be read as precise. Second, **the large point estimates are the imprecise ones:** 3.83 (2.63–5.57) for smoke-free-home adoption and 2.51 (1.62–3.88) for acute stress, while the tightest interval on the axis (1.33, 1.32–1.34, Korean incident asthma) belongs to one of the smallest effects and comes from 8.25 million people. Third, several estimates here are *co-pollutant* rather than PM — ozone, NO_2 , SO_2 — and are plotted because the papers report them alongside PM; read the labels before reading the positions.

Life-course windows addressed this period (records may span several stages)

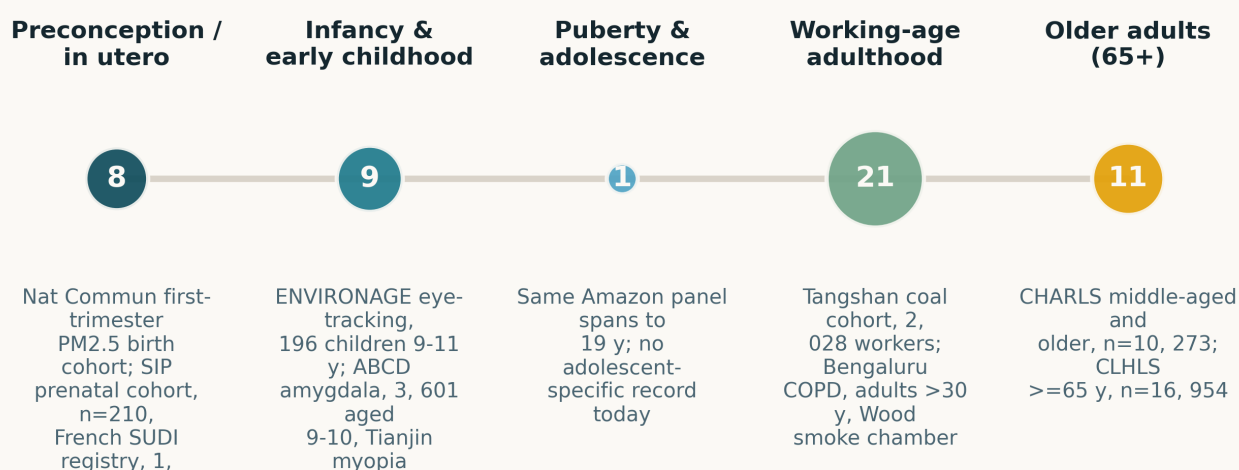


Fig. Life-course windows across the pooled week. Working-age adulthood dominates as usual; the change against last week is childhood, lifted by two measurement-side records (schoolchild personal monitoring, a Dutch school air-cleaner trial) rather than by epidemiology.

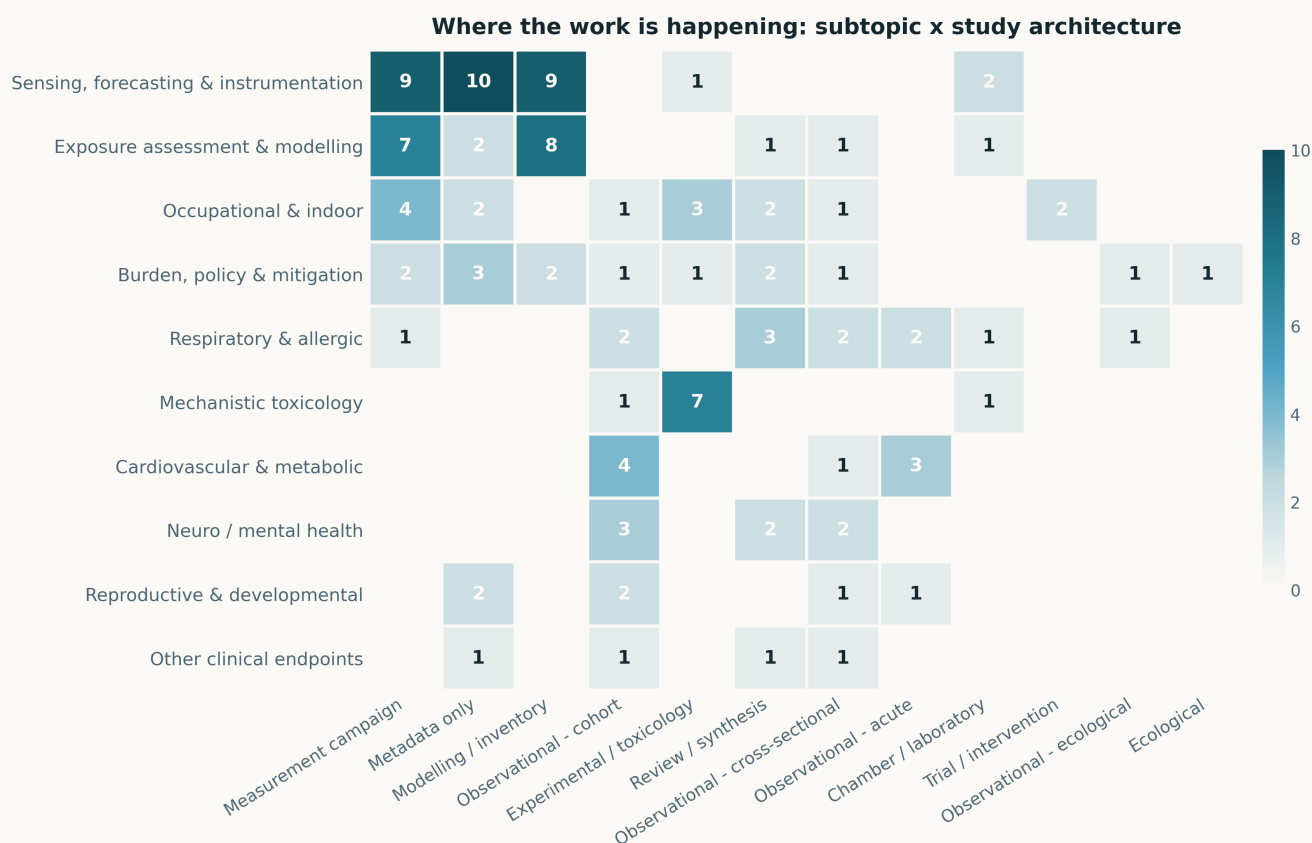


Fig. Subtopic $\times$ architecture, pooled into 51 cells. The densest cell in the entire week is *Sensing* $\times$ *Metadata only* at 10 — ahead of *Sensing* $\times$ *Measurement campaign* (9) and *Sensing* $\times$ *Modelling / inventory* (9). The standing structural finding survives a seventh week: of the 51 records in the *Sensing* and *Exposure* rows, 5 carry any human health endpoint. Instrumentation and epidemiology are still being written by different people in different journals, and this week the instrumentation half arrived without abstracts.

Venues, trial watch, and what the pipeline did

Venues. Far less concentrated than last week: *Environ Pollut* **10 of 126** leads, against *Environ Sci Technol*'s 23 of 150 a week ago. *Front Public Health* (7), *Environ Res* (6) and *Atmos Chem Phys* (6) follow, then *Atmosphere* (5) and *J Aerosol Sci* (4). The venue table is a harvesting artefact as much as an editorial one — titles on the Crossref by-ISSN list enter at their own publication rate, while journals reached only through PubMed enter at PubMed's cadence, and this week PubMed's cadence included two near-dead weekend days. **Preprints supplied 8 records** — 4 Research Square, 4 Preprints.org — all labelled as such in the register.

Recurring authors. Unchanged and unresolvable: surname clustering is dominated by high-frequency Chinese surnames, which is a naming artefact rather than a research-group signal, and this watch still has no author-disambiguation or affiliation field. **No repeat-group cluster is claimed for this week.** Adding a first-author affiliation field remains the open item that would make this section mean anything; it has been on the list since 3 September and is now the oldest unaddressed schema gap.

Trial watch — a genuine zero week, the first this watch has recorded. ClinicalTrials.gov returned **0 PM or air-pollution registrations or updates across 6–12 September**, against **4,413 studies updated registry-wide** in the same window. That control matters because the 5 September rollup established that weekend zeros are *structural* — the registry posts nothing on Saturdays and Sundays — and could not, on weekend data alone, distinguish a quiet registry from a quiet field. A seven-day window containing five active weekdays and 4,413 registry-wide updates removes that ambiguity: **this is a topical zero.** Against one new interventional registration and two updates last week, it is a real if unremarkable fluctuation, and one week of it means nothing on its own. `trials.json` still holds **26 trials**, now across **12 windows**. `analyze_endpoints` was not run — there was nothing to run it on.

What the harvesting legs did. Crossref-by-ISSN was the week's **workhorse and its liability**, carrying the instrumentation literature PubMed cannot see and delivering 10 of its *Sensing* records without abstracts. **PubMed** carried five of seven days and was effectively absent on both weekend days — 26 records registry-wide on 12 September is the lowest single-day figure this watch has measured, and the 6 September Sunday behaved the same way. **Europe PMC** was the quiet success, returning 17 on 12 September alone including two Research Square preprints, and it is now the leg that keeps weekend issues from collapsing. **OpenAlex returned HTTP 429 on every day of the week**, unchanged since 28 July, and remains a dead leg. **arXiv** returned 429 or nothing in window all week and contributed no record to any issue in this period — second consecutive week of that, and the query is now overdue for inspection. **Consensus returned a nil every day**, and from 9 September onward the nil has been verified the correct way, on Crossref `created` date rather than publication date; its hits are consistently the right literature with dates years outside the window.

Open gaps

- ▶ **Twenty records this week assert no finding.** 16% of the corpus is title-only, half of it in *Sensing*. This is the single largest recoverable loss in the pipeline and the fix is mechanical: a scheduled retry pass over the standing abstract-retry list, re-querying Crossref per DOI, Europe PMC and PubMed a week after first sight, when the publisher deposit has landed. Nothing about it requires new sources.
- ▶ **Five of 51 measurement-side records carry a health endpoint**, for the seventh consecutive week. This remains the longest-standing structural finding in the series, and it is now supported by 51 records in a single week rather than by a handful.
- ▶ **Ultrafine and size-resolved number: 3 of 126.** The strongest sensing record of the week — a community CPC matching a research-grade reference at $R^2 > 0.99$ with a 3% inter-unit coefficient of variation — exists precisely to make that metric affordable. The instrument is arriving faster than the studies that would use it.
- ▶ **Two interventional records in 126**, and neither tests a monitoring technology. The engineering turn in the indoor cluster is still being evaluated against chambers and models rather than against health outcomes in occupied buildings.
- ▶ **Pipeline defects carried forward:** unnormalised geography labels (which inflate the *Global / multi-region* bar in f3), the *Cancer/Onco*logic split in `ENDPOINT_CANON`, no first-author affiliation field, and a 6-element `LIFECOURSE` array in the 2026-08-03 corpus where the schema requires 5 — outside this window, but it will corrupt any rollup that spans it, including the September monthly.

Paper-level digest — the week by subtopic

Every record that appeared in a daily issue keeps that issue's critical summary verbatim; nothing here is re-written or re-assessed. Records surfaced without an abstract are shown as such. Ordered by subtopic, then by the date of the issue that first carried the record.

Sensing, forecasting & instrumentation

31 records

CARRIED FROM THE DAILY ISSUES — FULL SUMMARIES (31 OF 31 RECORDS IN THIS SUBTOPIC)

3D-LSS particle shape 2026 J Aerosol Sci

METADATA ONLY — *enhanced 3D light-scattering sensor for online particle shape*

No abstract retrievable on the entry date. The title commits to two things: an *enhanced* three-dimensional light-scattering sensor for online aerosol shape characterisation, and a validation against test aerosols. Neither the size range, the angular sampling scheme, the sphericity metric nor the validation particle set is recoverable, and none is asserted here. Shape is the property optical low-cost sensors silently assume away when they convert scattering to mass, so an online shape channel is directly relevant; the predecessor 3D-LSS work established the concept at 500 nm–5 μm , and what “enhanced” means against that baseline is precisely what the abstract would settle.

doi: 10.1016/j.jaerosci.2026.106860

Aerosol core/morphology optics 2026 J Aerosol Sci

Coupled Effects of Core Material and Aggregate Morphology on the Optical Properties of Atmospheric Aerosols

METADATA ONLY. Metadata only; no abstract in Crossref, Europe PMC or PubMed on per-DOI retry. Retained on title because core composition and aggregate morphology set the scattering cross-section that every optical PM sensor inverts to mass, which is the physical root of the humidity and composition dependence seen in co-location studies. No finding is asserted. On the retry list.

doi: 10.1016/j.jaerosci.2026.106897

Alkylamine IC-MS/MS method 2026 J Chromatogr A

Speciated atmospheric alkylamines at picogram levels in gas and particle phase

Alkylamines drive new particle formation and early aerosol growth, and are analytically awkward — strongly polar, present at trace level, prone to adsorption and decomposition. This method pairs alkaline purging with acidic absorption for selective enrichment ahead of ion chromatography–tandem MS. Method detection limits are **9–94 pg m^{-3} gaseous and 0.1–1.1 pg m^{-3} in PM**, a substantial gain on existing offline methods, and the approach resolves the dimethylamine/monoethylamine isobaric pair and quantifies tertiary amines that derivatisation-based methods structurally cannot. Field-applied at septic tanks and traffic tunnels in Dalian, it recovered a broader speciation profile than previous work and seasonal variation in each species. It is an offline enrichment method, so time resolution is whatever the sampling interval is — this does not compete with online amine measurement for process studies, it competes on speciation and sensitivity.

doi: 10.1016/j.chroma.2026.467416

BC coating optics 2026 J Aerosol Sci

METADATA ONLY — *metal-oxide coatings and the optical properties of black carbon*

No abstract retrievable on the entry date. The record establishes that the optical effect of V_2O_5 , NiO and Fe_2O_3 surface coatings on black carbon aggregates has been characterised; it establishes no absorption enhancement factor, wavelength dependence, coating thickness or morphology treatment, and none is asserted here. The reason to track it: absorption-based BC measurement — aethalometers, photoacoustic instruments and the filter-based methods most networks run — is systematically biased by the coating enhancement factor, and the metal-oxide coatings specifically are the industrial and brake-wear case rather than the usual organic or sulfate one.

doi: 10.1016/j.jaerosci.2026.106885

Baltic ship plume OFR aging 2026 Environ Sci Technol Lett

Fresh and photochemically aged emission factors from individual ships, measured at sea

An oxidation flow reactor operated from a moving research vessel, sampling identified ocean-going ships at **0.6–1.6 km** standoff in the Baltic, plus the research vessel's own plume as a control. Gas-phase and potential-aerosol-mass emission factors were measured fresh and after photochemical aging. **Particle mass rose after aging for every vessel sampled**, and secondary aerosol formation potential scaled with SO_2 emission — i.e. with fuel sulfur content, consistent with sulfate-driven particle formation. This is a pilot with a small vessel sample and no size-resolved chemistry reported in the abstract, so the emission factors are demonstrative rather than inventory-grade. The contribution is the framework: stack-level aging experiments and fixed ambient monitoring have both existed for years, and this joins them on a platform that can follow an individual, identifiable source.

doi: 10.1021/acs.estlett.6c00695

Bogota BiLSTM Huber 2026 Atmosphere

Asymmetric-loss BiLSTM for 24 h $\text{PM}_{2.5}$ forecasting and exceedance detection

Hourly records from the Puente Aranda industrial-corridor station, Bogotá, 2022–2025 ($n = 31,961$). The motivating statistic is sharp: **98.7%** of days exceeding the WHO 2021 $15 \mu\text{g}/\text{m}^3$ guideline contain at least one hour above 25, median peak-to-mean **1.90**, so a model trained on mean error systematically misses the episodes that matter. Three mechanisms are stacked — an asymmetric Huber loss penalising underprediction α -fold, exceedance sample weighting, and Optuna-TPE over 100 trials. The honest result is a reported **regression–detection tension**: no α minimised both global and peak MAE. Optimisation chose $\alpha = 2.80$, giving the best detection (AP **0.724**, 95% CI 0.582–0.821) and lowest peak MAE ($10.64 \mu\text{g}/\text{m}^3$) at the cost of $R^2 \approx 0$ globally; $\alpha = 1.5$ is the compromise (AP 0.718, RMSE 6.65, $R^2 = 0.372$). One station, so nothing here is a spatial claim, and the wide bootstrap intervals on AP overlap across all configurations — the ranking is weaker than the point estimates suggest.

doi: 10.3390/atmos17090871

Boston sparse network 2026 Environ Sci Technol

What sparse fixed air-quality networks miss, measured against mobile monitoring

Fixed-station $\text{PM}_{2.5}$ paired with simultaneous mobile measurement at a dense set of locations within a 250 m radius, at three contrasting areas of the Boston metro area across spring and summer 2025. Typical agreement is close — mean difference **$2.0 \mu\text{g}/\text{m}^3$** , median 1.5, IQR 0.6–2.8 — but dynamic events drove differences past 300% and **$800 \mu\text{g}/\text{m}^3$** , and **10.4% of cases** (21.8% at one site in spring) fell either side of an AQI category boundary. Crucially **Euclidean distance alone could not explain the discrepancies**, which undercuts radius-based representativeness and every inverse-distance product built on a reference network. Relative humidity was the strongest reported correlate ($r = -0.84$), a signed rather than absolute association that the abstract does not fully disambiguate. Two seasons and one metro area, and mobile platforms carry inlet and response-time artefacts of their own; but this is the number the dense-network argument has been missing.

doi: 10.1021/acs.est.6c03923

CALIOP Shanghai aerosol stratification 2026 Atmos Res*Long-term CALIOP observations of vertical aerosol stratification over Shanghai: Meteorological modulation and height-resolved transport*

METADATA ONLY. Metadata only - no abstract deposited to Crossref and none in Europe PMC or PubMed on per-DOI retry. Carried on title because height-resolved spaceborne lidar climatology over a megacity is the long-record complement to the single-tower fusion result in this same issue, and because column-to-surface conversion error is the central obstacle for satellite PM_{2.5}. No finding asserted. On the retry list.

doi: 10.1016/j.atmosres.2026.109329

Climate-stratified PM_{2.5} LightGBM 2026 PLoS One*Climate-stratified assessment of PM_{2.5} using machine learning: Geographic controls dominate meteorological factors*

49,997 monthly MERRA-2 samples (2019-2023) across five latitude-band climate zones under Leave-One-Climate-Out cross-validation. Tuned LightGBM reaches a global LOCO R² of 0.367 (MAE 6.43 $\mu\text{g m}^{-3}$), but zone predictability ranges from R² 0.604 subtropical to 0.142 tropical. SHAP puts 64% of importance on geographic coordinates against 23% on meteorology, and a retrain ablation confirms the coordinates are not mere spatial autocorrelation - dropping lat/lon collapses mean R² from 0.367 to -0.099. Temporal split validation gives 0.544, so LOCO is the harder test. The honest limitation is that the target is reanalysis PM_{2.5}, not measured PM_{2.5}, so the model is benchmarked against a model; and coordinates proxying emissions means the framework cannot transfer to a region whose emission geography it never saw.

doi: 10.1371/journal.pone.0347591

Cooking aerosol FIGAERO partitioning 2026 Atmos Chem Phys*Unravelling gas-particle partitioning dynamics in cooking aerosol oxidation through FIGAERO-CIMS analysis*

Heated corn oil emissions aged in a Go:PAM oxidation flow reactor, with FIGAERO-I-CIMS and a 2-D VBS parameterisation tracking composition, volatility and partitioning. Kinetic calculations give molar-mass-dependent partitioning with the surface process rate-limiting. Primary cooking aerosol sits biased toward the gas phase relative to equilibrium; in secondary aerosol the crossover is near 120-220 g mol^{-1} , below which compounds favour gas and above which they favour particle. The honest result is the negative one: a substantial unresolved gap remains between FIGAERO-derived effective volatility and equilibrium thermodynamic estimates, which the authors attribute to combined measurement and model uncertainty rather than resolving. Cooking is a rising share of urban OA as combustion sources are controlled, and non-equilibrium partitioning is a direct problem for any mass-based sensor calibrated on ambient mix

doi: 10.5194/acp-26-12793-2026

Coriolis sampler inactivation 2026 Indoor Air*Sampler-induced loss of respiratory virus infectivity in spiked collection liquid*

Rhinovirus, SARS-CoV-2 and influenza A spiked into the collection liquid of a Coriolis μ liquid impinger and run at several airflow rates — deliberately *not* an aerosol or ambient-air collection, which isolates mechanical stress in the liquid from capture efficiency. Culturable virus fell by up to 98.6% for SARS-CoV-2 and influenza A, while RNA copies were largely unchanged for SARS-CoV-2 and rhinovirus and reduced only for influenza A. Rhinovirus infectivity was essentially unaffected, so the effect is virus-specific rather than a general artefact. The design's strength is also its limit: it cannot say how much infectivity is lost during aerosol capture itself, only during the liquid phase, so it bounds one term of the total sampling loss. The consequence for bioaerosol surveillance is direct — a PCR-positive, culture-negative air sample is not evidence of non-infectious aerosol.

doi: 10.1155/ina/5068026

Cyprus dust typing + DRE 2026 Atmos Meas Tech*Characterization of Saharan and Middle Eastern dust aerosols and their associated shortwave direct radiative effects over Cyprus*

Seven years (2015-2022) at Agia Marina Xyliatou. Dust is 28.3% of aerosol cases in MAM and 12.7% in SON; 86% Saharan, 14% Middle Eastern. AOD₄₄₀ 0.34 $\pm$ 0.13 Saharan vs 0.40 $\pm$ 0.13 Middle Eastern, SSA₄₄₀ 0.93 for both, i.e. the two source regions are separable by trajectory and loading but not by single-scattering albedo. Modelled GHI within $\pm$ 5% of ground irradiance for 87% of cases. At SZA 60 deg the strongest monthly mean surface forcing is -92 W m^{-2} (Saharan) and -86 W m^{-2} (Middle Eastern), with +72 and +59 W m^{-2} of atmospheric heating. Limitation: DRE is reported at one fixed SZA, so the monthly means are not diurnally integrated and cannot be read as energy budgets. Column optical typing is the reference against which a low-cost PM network's dust-episode flags should be scored

doi: 10.5194/amt-19-5753-2026

Europe BLH regression kriging 2026 Atmos Meas Tech*Monthly boundary-layer height maps for Europe at 25 km from radiosonde, satellite and ERA5*

Regression kriging over 2010-2020 combining radiosonde profiles with land-surface temperature, topographic covariates and ERA5, at 25 km and monthly means for 00 and 12 UTC. Random forest beat linear regression and gradient boosting at both hours. Against ERA5 the product is consistently sharper: winter spatial variability at 12 UTC gives RMSE $\leq$ 170 m versus $\geq$ 250 m, summer temporal variability $\leq$ 260 m versus 340 m. Performance declines considerably at 00 UTC, which is the honest limitation and the awkward one — the nocturnal stable layer is exactly where mixing-height error propagates hardest into surface concentration. Monthly means also cannot serve episode-scale exposure work. Still, boundary-layer height is the dominant meteorological term in almost every surface PM model, and a validated European product that beats reanalysis is directly usable as a covariate.

doi: 10.5194/amt-19-5697-2026

FY-4B/AGRI AOD evaluation 2026 Atmos Environ*Evaluation of the FY-4B/AGRI Land Aerosol Optical Depth Product and Comparison with Geostationary Satellite Products over Asia*

METADATA ONLY. No abstract was available on the entry date from any of the four sources queried - the same same-day Elsevier deposit lag logged repeatedly since 2026-09-08. Title and journal establish a validation of the FY-4B/AGRI land AOD product against other geostationary retrievals over Asia. No validation statistics, no reference network and no spatial or temporal coverage are available, so nothing is asserted here and the record is tier C. Worth re-reading when the abstract deposits: geostationary AOD is the only route to sub-daily satellite-surface PM fusion, and product-versus-product agreement is the term that dominates the error budget of any such fusion

doi: 10.1016/j.atmosenv.2026.122354

Genoa absorption intercomparison 2026 Atmosphere*AE33 aethalometer against two filter-based absorption techniques in a coastal city*

Multi-wavelength field intercomparison at the Multedo urban coastal site, Genoa, where the aerosol mixes traffic, shipping, industry and marine air. The AE33 was run against the Multi-Wavelength Absorption Analyzer and the Broadband Light Analyzer of Complex Aerosols across UV, visible and near-IR. Correlations are excellent — R^2 0.93-0.99, tightest between the two offline methods — but AE33 reports absorption coefficients 12-22% higher, with the offset varying by wavelength. That pattern implicates the instrument's default multiple-scattering correction, which is tuned to a generic aerosol and not to this one. Absorption Ångström exponents from all three sit near unity, consistent with dominant primary combustion. The limitation is that one site and one campaign cannot fix the correction factor generally — which is precisely the point: a site-specific bias is not a constant you can publish once.

doi: 10.3390/atmos17090883

India winter PM_{2.5} seasonal forecast 2026 Sci Adv*Winter PM_{2.5} severity over northern India is predictable a season ahead*

Over a decade of quality-controlled Indian government reference PM_{2.5} monitoring, used to establish that **roughly 70% of interannual variance** in northern Indian winter PM_{2.5} is explained by western disturbances — Mediterranean-origin extratropical storms — acting through ventilation and wet removal. The storms in turn track to the position and strength of the subtropical jet, which is traceable to North Atlantic and tropical Indian Ocean sea-surface-temperature anomalies in the preceding autumn. Using only pre-season SST, the authors predict winter WD precipitation and PM_{2.5} severity one season ahead at r^2 **0.52–0.69**. The honest caveat is that this is meteorological predictability, not emissions predictability: it forecasts how well a given emission load will be ventilated, and a decade of records is a short sample for an interannual teleconnection. For network planning it is nonetheless actionable — it says which winters to instrument heavily.

doi: 10.1126/sciadv.aee4549

LCS field calibration correction 2026 J Environ Health Sci Eng*Correction: Long-term performance evaluation and seasonal field calibration of a low-cost particulate matter sensor*

METADATA ONLY. Correction notice to 10.1007/s40201-026-00986-0. The notice carries no content of its own; logged here only so the archive records that the underlying long-term low-cost-sensor calibration paper has been amended, and so a later reader of that record checks which figures or coefficients moved before reusing them. No finding is asserted and the record is tier C on that basis

doi: 10.1007/s40201-026-01003-0

NaCl aerosol turbulent size redistribution 2026 J Aerosol Sci*Turbulence-Induced Size Redistribution of Submicron NaCl Aerosols in Ribbed Confined Flows Prior to Electrostatic Collection*

METADATA ONLY. Metadata only. Crossref deposited this DOI on 12 September with no abstract, and per-DOI retries against Crossref, Europe PMC and PubMed all returned nothing - the expected same-day Elsevier deposit gap. Carried on title alone because submicron size redistribution in confined turbulent flow directly governs what reaches an inlet, and no finding is asserted here. On the retry list for a later issue.

doi: 10.1016/j.jaerosci.2026.106899

Open-pit mine dust stacking model 2026 Atmos Pollut Res*METADATA ONLY — multi-point cold-season dust prediction at an open-pit coal mine*

No abstract existed in Crossref queried per-DOI, in Europe PMC or in PubMed on the entry date. What the record establishes: a multi-point dust-concentration prediction task at an open-pit coal mine during the cold season has been addressed with a stacking ensemble and an interpretability analysis. What it does not establish: the number and siting of monitoring points, the prediction horizon, the base learners in the stack, the interpretability method, any error figure, or the mine's location — none of which is inferable from the title and none of which is asserted here. Flagged for retrieval; open-pit mining dust is one of the few settings where a dense low-cost network has an unambiguous operational customer.

doi: 10.1016/j.apr.2026.103196

PPDRTL leaf-image AQI 2026 PLoS One*Predicting roadside air quality from image texture of pollutant deposits on tree leaves*

High-resolution images of roadside leaves at high, medium and low traffic-density sites in Haryana, segmented with PSO, DPSO and FODPSO, reduced to contrast, entropy and standard deviation features and regressed with ordinary linear regression onto Haryana State Pollution Control Board reference AQI. The FODPSO variant reaches R^2 **up to 0.894** with **RMSE 8.76–12.34 $\mu\text{g m}^{-3}$** and reported site accuracies of 88.9–91.4%. The inference is weak for a reason worth naming precisely: the **pilot dataset is ten days**, no held-out test set is described, and leaf deposition is a *cumulative* integrator being regressed onto an *instantaneous* daily index, so the model is fitting a mismatched target. Mixing an R^2 against a unitless AQI with an RMSE in $\mu\text{g m}^{-3}$ compounds the confusion. Filed as a proxy-sensing curiosity, not a method.

doi: 10.1371/journal.pone.0326588

RI-URBANS urban AQ mapping 2026 Atmos Meas Tech*What a continent of low-cost-sensor and citizen-science air-quality pilots actually taught*

Synthesis of the RI-URBANS mapping and hotspot-identification pilots across European cities, covering stationary low-cost sensor networks, mobile monitoring platforms and citizen-collected data alongside reference networks and modelling. The conclusion the programme reaches is organisational rather than technical: the limiting factor on producing comparable urban maps was **harmonisation between methods**, not the accuracy of any one method, and harmonisation was one of the project's stated primary aims for that reason. Two operational findings are worth carrying into any deployment — campaign design and sustained, direct communication with citizen participants determined data quality as much as hardware did. As a programme-level synthesis it reports no single effect estimate and cannot be appraised like a primary study; its value is the negative result about where the effort goes.

doi: 10.5194/amt-19-5729-2026

SAQIN 2026 Neural Netw*Long-short-term decoupled residual learning for high-resolution spatio-temporal air quality inference*

Inference — not forecasting — of high-resolution concentration fields from sparse reference stations. The model splits a static branch (semantic segmentation, remote-sensing context, population density, elevation) from a dynamic branch (multi-head self-attention over geographic relationships and inter-pollutant coupling), then treats inference as residual correction anchored to multiple stations, fused by distance-weighted averaging across stations. That last step is the substantive contribution: it removes the Voronoi-partition discontinuities that single-anchor schemes produce, yielding a continuous field. Reported as superior on large-scale real-world data with zero-shot cross-domain generalisation, but the abstract names no dataset, no station count, no RMSE and no baseline, so none of it is checkable here. The architecture is directly applicable to dense low-cost networks, where the anchor stations are the few reference-grade nodes.

doi: 10.1016/j.neunet.2026.109575

SE India monsoon aerosol 2026 Atmosphere*Seasonal aerosol optical properties and radiative effect at a rural Indian receptor site*

MICROTOS-II sunphotometry from April 2021 to December 2023 at a rural site in Southeast India, with trajectory-based source attribution (PSCE, CWT) and OPAC-SBDART radiative transfer. Annual mean AOD₅₀₀ **0.56 ± 0.22**, peaking pre-monsoon (0.66 ± 0.19) and in winter (0.64 ± 0.23) and falling to 0.49 ± 0.21 in the monsoon. **Angstrom exponent 1.30 ± 0.24** in the dry seasons against **0.83 ± 0.37** in the monsoon marks the fine-to-coarse shift, corroborated by negative spectral curvature. Precipitable water rose 2.00 to 4.35 cm winter to monsoon, which confounds the optical size retrieval through hygroscopic growth — the paper says so. Column optical depth is not surface mass, and a single rural receptor cannot be generalised; its value is as a South Asian background record against which urban surface networks can be referenced.

doi: 10.3390/atmos17090876

SIBS inexpensive single-particle 2026 Aerosol Sci TechnolMETADATA ONLY — *feasibility of a low-cost single-particle SIBS instrument*

No abstract existed in Crossref, Europe PMC, Semantic Scholar or on the publisher page on the entry date, and Taylor & Francis refused the fetch. What the record establishes: a feasibility study for an *inexpensive* single-particle spark-induced-breakdown-spectroscopy instrument has been published in the field's instrumentation journal of record. What it does not establish: any detection limit, element set, particle-size range, spark energy, hit rate or cost figure — none of which can be inferred from the title, and none of which is asserted here. This is the most directly relevant record of the day to low-cost elemental speciation and is flagged for retrieval once the abstract deposits.

doi: 10.1080/02786826.2026.2721653

Shenzhen tower vertical PM_{2.5} DL 2026 J Hazard Mater*Leveraging multi-source observations for vertical PM_{2.5} estimation: A deep learning fusion model*

An AM-CNN-BiLSTM fusion model estimating PM_{2.5} at 200 and 300 m on the Shenzhen Meteorological Gradient Tower, July 2022-June 2023, combining lidar aerosol extinction coefficient, microwave-radiometer thermodynamic profiles, meteorology and tower PM_{2.5}. Chronological test split with an expanding window gave R² 0.87 at 200 m and 0.83 at 300 m, beating baselines; SHAP, ablation and sensitivity analysis all identify extinction coefficient as the dominant constraint, with RH, temperature and wind complementary. The authors state plainly that the learned extinction-PM_{2.5} and humidity relations are site-specific and need multi-year cross-site validation - with one tower and one year that caveat is binding. Still the cleanest demonstration that lidar plus profiler can substitute for vertical in-situ nodes.

doi: 10.1016/j.jhazmat.2026.143445

TROPOMI SO₂ EAKF assimilation 2026 Atmosphere*Assimilating TROPOMI SO₂ columns into a regional chemical transport model*

An Ensemble Adjustment Kalman Filter implemented in the MINNI regional CTM through DART, assimilating Sentinel-5P TROPOMI SO₂-COBRA total columns over continental Europe for August 2023, with a 20-member ensemble perturbing emissions and boundary conditions. The filter constrained ensemble variance around power-plant and volcanic plumes; the monthly domain-mean column correction was $2 \times 10^{-5} \text{ mol m}^{-2}$ with localised maxima of 3.3×10^{-4} , and near-surface concentration corrections reached $2.6 \mu\text{g m}^{-3}$, validated against in-situ data. The stated limitations are real ones: small ensemble, static vertical localisation, and the usual temporal fading of initial-condition corrections. Relevance here is indirect but concrete — SO₂ is a sulfate precursor, and column-to-surface transfer skill is the same problem satellite-based PM_{2.5} products have to solve.

doi: 10.3390/atmos17090875

UAV single-particle CCSEM 2026 Environ Sci Technol*Size-resolved aerosol composition and mixing state through the boundary layer from a UAV*

A microscopy-based particle sampler flown on a UAV at up to 30 m vertical and 1 min temporal resolution in a coastal boundary layer; over **100,000 particles** classified by computer-controlled SEM with EDX. The observed transition runs from sea-breeze aerosol dominated by aged sea salt and dust (**46–83%**) to a continentally influenced population dominated by organic-sulfur particles (**57–89%**), and the mixing state index falls from **53% to 27%** as those particles form — i.e. toward external mixing. Mixing state showed strong diurnal and size dependence but weak vertical structure. Offline microscopy is not a real-time measurement and the sampling is necessarily intermittent, so the “temporal evolution” is a sequence of snapshots; one coastal site, and no companion mass measurement is described. The transferable point for network design is that composition and mixing state vary on scales a fixed filter sampler averages away entirely.

doi: 10.1021/acs.est.6c03627

US precipitation washout DoubleML 2026 Research Square*Observation-derived washout coefficients, and how wildfire smoke breaks them*

Preprint. Ten years of hourly EPA observations — 760 PM_{2.5} and 451 PM₁₀ monitors across the contiguous US — merged with NLDAS-2 meteorology and NOAA HMS smoke classifications, with a double machine-learning causal framework separating precipitation from confounding meteorology. Removal rises with rainfall intensity, particle size and initial concentration, and **falls as wildfire smoke loading rises** — smoke aerosol is less efficiently scavenged, which matters more each year. The authors fit power-law washout coefficients ($a = 0.0404$, $b = 0.5951$ for PM_{2.5}; $a = 0.0812$, $b = 0.4508$ for PM₁₀) intended to drop straight into chemical transport models. The inference threat is the usual one for DoubleML on observational data: unmeasured confounding is assumed away rather than eliminated, and hourly monitor-scale precipitation from a reanalysis is a coarse proxy for what fell on the sampler.

doi: 10.21203/rs.3.rs-10855101/v1

W Sub-Saharan Africa PM monitoring 2026 Atmos Pollut ResMETADATA ONLY — *PM_{2.5} and PM₁₀ monitoring capacity in Western Sub-Saharan Africa*

No abstract retrievable on the entry date. The record establishes that a challenges-and-perspectives treatment of PM_{2.5} and PM₁₀ monitoring and estimation across Western Sub-Saharan Africa has been published; whether it is a systematic review, a narrative synthesis or a data-driven assessment cannot be determined from the metadata, and no finding is asserted here. Western Sub-Saharan Africa has the sparsest reference network of any populated region and is the canonical justification for low-cost sensor deployment, so a current capacity assessment is worth retrieving — particularly alongside the Accra hotspot record in this same issue.

doi: 10.1016/j.apr.2026.103195

Wildfire CHON PM_{2.5} 2026 Environ Res*FT-ICR MS with rule-based prediction and QSAR for nitrogen-containing organics in wildfire PM_{2.5}*

Molecular-level characterisation of PM_{2.5} from a temperate coniferous forest wildfire in South Korea, wildfire versus post-wildfire periods. GC-MS gave nitrated polycyclic aromatic compounds **3.4× higher during the fire (0.122 vs 0.036 ng/m³)**, with 2-nitrofluorene then 3-nitrofluoranthene dominant; FT-ICR MS showed a relative shift toward CHON formulae, and rule-based SOA prediction from 12 VOC precursors matched limonene-derived products most often. QSAR LC₅₀ values of 1.0, 1.6 and 8.3 mg/L for β -pinene-, α -pinene- and isoprene-derived compounds are ranked hazard, not measured toxicity, and the authors say so — the assignments are formula-level, so isomers with very different toxicity are pooled. One fire, one forest type; the compositional shift is the result, the hazard ranking is a hypothesis.

doi: 10.1016/j.envres.2026.125576

cCPC/OPC/SPS30 field evaluation 2026 Aerosol Sci Technol*Field Evaluation of Community Condensation Particle Counters, Optical Particle Counters, and a Low-Cost Particle Sensor*

Three community CPCs, three OPCs and one TSI BlueSky (Sensirion SPS30) co-located at an urban regulatory station for about four months, data recovery above 98%. Against a research-grade reference CPC the cCPCs gave R² above 0.99 with concentration ratios near unity and a 3% coefficient of variation across the three units - genuinely reference-comparable ultrafine number at community cost. OPC fine and coarse volumes correlated with reference PM_{2.5} and PM_{10-2.5} but on different regression slopes, so a single mass factor cannot serve both fractions. The SPS30 tracked PM_{2.5} and failed on coarse mass, which is the known optical limit restated with numbers. One season at one site, so no drift or humidity-regime generalisation; this is the most directly usable record of the day for network design.

doi: 10.1080/02786826.2025.2590103

Exposure assessment & modelling

20 records

CARRIED FROM THE DAILY ISSUES — FULL SUMMARIES (20 OF 20 RECORDS IN THIS SUBTOPIC)

Accra traffic intersections 2026 Environ Monit Assess*Meteorological drivers of PM_{2.5} at nine Accra traffic hotspots*

An Aeroqual Ranger portable monitor at nine traffic intersections in Accra across the harmattan and non-harmattan seasons, against Ghana Meteorological Agency station data. Median PM_{2.5} was **54.90 $\mu\text{g m}^{-3}$** at high-density sites, **34.26** at commercial and **16.75** at low-density — the cleanest category still exceeds the WHO 24 h guideline of 15. Temperature correlated positively ($r = 0.353$, $p < 0.01$); rainfall, wind speed and humidity negatively outside the harmattan, while in the harmattan wind speed flipped weakly positive ($r = 0.079$, $p = 0.450$, i.e. null) consistent with dust resuspension. The analysis is Pearson correlation only, with no multivariable adjustment and no traffic counts, so “the dominant factor remains localised vehicular emissions” is a residual argument rather than a measured one. A single portable instrument across nine sites also gives no co-location error term.

doi: 10.1007/s10661-026-15889-8

Benin PM_{2.5}-water grid screening 2026 Environ Monit Assess*Grid-based screening of fine particulate matter and water-optical conditions in Southern Benin: the Cotonou, Lake Nokoue, and Porto-Novo Corridor*

A 1 km screening framework over 2,589 grid cells in the Cotonou-Lake Nokoue-Porto-Novo corridor, 2001-2022. Mean PM_{2.5} was 32.17 $\mu\text{g m}^{-3}$ for 2001-2022 and 32.70 $\mu\text{g m}^{-3}$ for 2018-2022, roughly six times the WHO annual guideline, with no corridor-wide trend surviving false-discovery-rate correction. Sentinel-2 NDTI tracked field turbidity (Spearman rho 0.488, 95% CI 0.369-0.590) and a red-edge ratio tracked chlorophyll-a (0.432, 0.295-0.544) against 19 stations. Six cells co-flagged at the 80th percentile, but the count swings from 16 at the 75th to zero at the 90th - the threshold, not the data, sets the answer. No ground PM monitor exists in the corridor, so the PM_{2.5} layer is wholly unvalidated: a concrete argument for low-cost reference-grade co-location in West Africa.

doi: 10.1007/s10661-026-15908-8

Brake wear SO₂ uptake 2026 Nat Commun*Unexpectedly rapid SO₂ heterogeneous reaction on the surface of vehicular brake wear particles*

Bench heterogeneous kinetics on collected brake wear particles. SO₂ uptake coefficient up to 2.87e-5 under dark ambient conditions, orders of magnitude above mineral dust or soot. The mechanism proposed is a self-sustained alpha-Fe₂O₃ / carbonaceous synergy: oxygen vacancies activate O₂ and H₂O to ROS and Fe-OH, while organics and elemental carbon promote H₂O dissociation and enhance SO₂ adsorption at carbon defects. Limitation: uptake coefficients are measured on collected, laboratory-handled particles under dry dark conditions, so the extrapolation to ambient RH and photochemistry is an assumption, not a measurement. Relevance: non-exhaust PM is now the dominant vehicular primary source in regulated fleets, and a reactive brake-wear pathway means roadside sulfate is not purely advected

doi: 10.1038/s41467-026-76820-w

Brisbane Aitken-mode PNC 2026 Environ Int*METADATA ONLY — long-term Aitken-mode particle number trends in Brisbane*

No abstract retrievable on the entry date despite *Environ Int* being a PubMed-indexed journal — the DOI indexed in Crossref before either the abstract or the PubMed record deposited. The title commits to a long-term change in Aitken-mode particle number concentration attributed across meteorology, traffic and biomass burning. Neither the record length, the trend sign or magnitude, the instrument, nor the attribution method is recoverable, and none is asserted here. Long records of sub-100 nm number concentration are rare and this one would bear directly on whether mass-based networks are tracking the health-relevant fraction; flagged for retrieval.

doi: 10.1016/j.envint.2026.110514

Construction equipment PM-bound EPFR 2026 Atmos Environ*METADATA ONLY — persistent free radicals on PM from non-road construction plant*

No abstract retrievable on the entry date from Crossref per-DOI, Europe PMC or PubMed. What the record establishes: PM-bound environmentally persistent free radicals and downstream reactive radical species have been measured from non-road construction equipment under *real-world* rather than dynamometer conditions, which is the harder and rarer measurement. What it does not establish: radical concentrations or lifetimes, the engine tiers and duty cycles sampled, the EPR or spin-trapping protocol, or the site — none is asserted here. Relevant because EPFR content is a redox-activity metric that mass concentration does not predict, and construction plant sits inside the urban domain that dense networks are deployed to characterise.

doi: 10.1016/j.atmosenv.2026.122353

Douala households 2026 Spatiotemporal Epidemiol*Particulate concentration and household respiratory health in urban Cameroon*

On-site campaign in Douala with an OC 300 Dust Particle Laser — a low-cost portable optical sensor — paired with a survey of 1,721 individuals in 344 households sited near the measurement points; instrumental Logit to address endogeneity of exposure. Reported effects on respiratory disease were **0.231 for PM₁₀**, **0.122 for PM₅** and **0.018 for PM_{2.5}**, and unsealed roads and traffic density dominated the PM levels. The size ordering is the problem: effect falls monotonically as particles get smaller, the reverse of what deposition physics predicts, and the most economical explanation is an uncorrected nephelometric instrument distributing mass across bins it cannot separate — no co-location against a reference monitor is described, and no humidity correction. Included because it is a rare health study built on a low-cost node, and because it shows precisely what the missing co-location arm costs.

doi: 10.1016/j.sste.2026.100836

EROD-HR/SOILHD Iberia 2026 Atmos Chem Phys*High-resolution soil erodibility and its effect on WRF-Chem dust and PM₁₀*

Two erodibility datasets tested in WRF-Chem with GOCART: EROD-HR, which folds fine-resolution topography in at 0.0625° globally and 1 km over Iberia, and SOILHD, which adds high-resolution sand/silt/clay texture and removes misclassified bare-soil areas at 1 km globally. Five dust episodes spanning 2022–2025 cover both local emission and long-range North African intrusion, evaluated against **PM₁₀ from the SINCLAIR network in Murcia**. Both datasets improved the magnitude and timing of modelled PM₁₀ peaks at inland stations; **improvements were limited at coastal stations**, which the paper does not fully resolve and which is the most interesting part of the result — coastal sites mix dust with sea salt and are the ones regulatory exceedance attribution most often disputes. Five episodes is a small evaluation set, and network PM₁₀ cannot separate dust from co-occurring anthropogenic mass.

doi: 10.5194/acp-26-12641-2026

GPS+LCS schoolchild exposure model 2026 Environ Pollut*Integrating GPS and Low-Cost Sensors to Enhance Personal PM_{2.5} Exposure Modeling for Schoolchildren*

918 person-days from northern Taiwan schoolchildren over one year with GPS-equipped portable PM_{2.5} sensors, pre-deployment calibrated against a Met One BAM-1020 (mean R² 0.996, slope 0.962). Personal exposure was dominated by within-subject variability (ICC 0.125); school was the highest-concentration microenvironment at 22.7 $\mu\text{g m}^{-3}$. The fully GPS-resolved microenvironment model reached R² 0.619 (74.0% accuracy) and a home-plus-school-only model 0.616 (71.7%) - so essentially all the gain comes from identifying two microenvironments, and full trajectory resolution adds nothing. That negative result is the useful one for network design: it sets a ceiling on what finer location data buys. Validation is internal against concurrent personal measurement; whether the models reduce exposure misclassification in health analysis is explicitly not tested.

doi: 10.1016/j.envpol.2026.129149

Hong Kong GEMA PM2.5 mobility 2026 Environ Res*1.7 million personal PM_{2.5} measurements against momentary stress and activity*

Two-day geographic ecological momentary assessment on 800 Hong Kong adults, integrating **1.7 million personal PM_{2.5} measurements**, 2.1 million accelerometer records, 4,800 momentary assessments and eye-level greenspace derived from 4.3 million street-view images, summarised over 5- to 600-minute windows and analysed across eight activity-travel contexts with a DAG-guided multi-relation double-ML framework. Higher PM_{2.5} tracked more light-intensity and less moderate-to-vigorous activity overall, and predicted worse *perceived* air quality; perceived pollution in turn predicted higher stress while measured PM_{2.5} did not. The result that should travel: **context-specific estimates ran in opposite directions** while cumulative associations with stress showed nothing sustained. Two days per participant is thin for a stress outcome and reverse causation between activity and exposure is unresolved, but the decomposition itself is the finding — a pooled null here is an artefact of aggregation.

doi: 10.1016/j.envres.2026.125659

Liupan tethered eBC 2026 Atmos Chem Phys*Vertical structure of carbonaceous aerosol over the Loess Plateau from tethered balloon*

Coordinated tethered-balloon profiling of carbonaceous aerosol and meteorology at Pingliang, eastern foothills of the Liupan Mountains, during July 2023 and July 2024 campaigns. Near-surface aerosols were **0.84 $\mu\text{g/m}^3$ eBC** and **1.24 $\mu\text{g/m}^3$ UVPM**, both decreasing with height. The substantive observation is that in the early morning and at night, when convection is weak, **UVPM aloft exceeded eBC** — attributed to nucleation from gaseous precursors rather than transport. Thermodynamic processes dominated vertical transport near the surface while enhanced dynamics aloft drove horizontal dispersion, and southerly air masses contributed disproportionately to UVPM. Two July campaigns at one site cannot separate seasonal from interannual variation, and eBC and UVPM are optical attributions from a filter-based absorption instrument, not chemical speciation — the nucleation interpretation is inference from the eBC/UVPM divergence, not a measured formation pathway.

doi: 10.5194/acp-26-12671-2026

Milan pizza-oven dispersion 2026 Preprints.org*Assessment of Air Quality Impacts from Wood-Fired Pizza Ovens: Comparison with Major Urban Emission Sources*

PREPRINT. Milan. Measured wood-fired-oven emission factors feed a dispersion model, with road traffic and domestic heating simulated on the same framework and all three compared against ARPA Lombardia monitors. Oven contributions concentrate in the city centre: 0.085–3.34 $\mu\text{g m}^{-3}$ of primary PM₁₀, only 0.0044–1.75 $\mu\text{g m}^{-3}$ of NO_x, but 19.70–34.28% of levoglucosan and 34.95–37.47% of benzo(a)pyrene. That split is the finding - a source that is minor by mass and major by tracer, which is exactly the pattern that breaks levoglucosan-based residential-wood-burning apportionment in a restaurant-dense centre. Limitations: preprint, single city, emission factors from a small appliance sample, and the ranges are model spreads rather than measured confidence intervals

doi: 10.20944/preprints202609.0792.v1

N China indoor Finf 48 homes 2026 Atmosphere*Determinants of indoor PM_{2.5} and infiltration factors across 48 dwellings*

Paired indoor/outdoor PM_{2.5} at 5 min in 48 residences across eight Northern Chinese cities, seven consecutive days in each of four seasons, **31,094 valid hourly observations** from 32,256 scheduled. A linear mixed-effects model with a residence random intercept and AR(1) residuals gives **marginal R² = 0.71, conditional R² = 0.91, ICC = 0.34**: dwelling identity carries a third of the variance that the fixed effects — outdoor PM_{2.5} (β = 0.872), window-open fraction (0.548), cooking (0.046), purifier-on fraction (−0.902) — cannot reach. Continuous purifier operation associates with **59.4% lower** indoor PM_{2.5}, which the authors correctly decline to call causal. Threat to inference: window and purifier use are self-reported and dominate the same model, and coal-heated Northern China is a specific regime. For distributed sensing this is the cleanest recent argument that per-dwelling measurement is not redundant with a good ambient field.

doi: 10.3390/atmos17090874

NZ tractor fleet PM inventory 2026 Sci Total Environ*An uncertainty-propagated PM_{2.5} inventory for an entirely unregulated source class*

Non-road mobile machinery is under-regulated and rising in relative importance as on-road sources are controlled; New Zealand is an instructive case because it pairs ultra-low-sulphur diesel and tightening on-road standards with *no* NRMM emission regulation at all. The inventory joins row-level registration microdata for 43,367 tractors, a licensing table defining 31,253 active units, and 29,278 dealer sales records (2017–2025), under an IPCC Tier 2 fuel-anchored framework with EMEP/EEA factors, tier hours calibrated against a CARB monitoring study, and fuel anchored to a 204 Ml national consumption target. 10,000-iteration Monte Carlo gives **115 t PM_{2.5} (95% UI 73–190)** and 2258 t NO_x. The distributional result is the useful one: **Gini 0.51 for PM_{2.5}, with the dirtiest 30% of tractors producing 69% of it**, and modernisation running at only +2.7 percentage points a year. Activity hours remain the weakest link — they are calibrated, not measured.

doi: 10.1016/j.scitotenv.2026.182302

Paediatric exposome 2026 Pediatr Res*Assessing environmental exposures in children: toward a comprehensive exposome approach*

Narrative review of exposure-assessment tooling for paediatrics — questionnaires, biomarkers, wearable and fixed sensors, geospatial indicators and pooled databases — alongside longitudinal cohort design, multi-omics integration and methods for mixtures and critical windows. No systematic search protocol, no pooled estimate, and the closing argument is professional advocacy rather than evidence. Useful only as a citation anchor for the claim that single-pollutant assessment is the limiting factor in child environmental epidemiology, and for its framing of sensors as one instrument among several rather than the exposure itself.

doi: 10.1038/s41390-026-05460-z

Paris Saharan dust heatwave 2026 Atmos Chem Phys*Dust aerosol cools Paris by up to 1 °C — and that does not mean comfort*

Three Meso-NH simulations of the 15–19 June 2022 Paris heatwave — no dust, CAMS reanalysis dust, and doubled dust — validated against the PANAME-Urban campaign, which the model reproduces for aerosol optical depth and incoming solar radiation, with improved air temperature and boundary-layer height once aerosol is included. The dust plume's radiative effect cuts surface shortwave by up to 75 W m^{-2} and air temperature by up to 1°C . The result worth the page is the one that refuses the easy conclusion: thermal comfort improves by up to 1°C in *sunlit* urban and suburban areas at 16:00 UTC, but in shaded suburban areas the increase in diffuse radiation and humidity plus the drop in wind speed cancels the temperature benefit entirely. A single five-day episode in one city, and comfort indices are not health outcomes — but the case for coupling aerosol forecasts to heat-health warning systems is made on the disaggregation, not on the mean.

doi: 10.5194/acp-26-12729-2026

St Elias proglacial dust 2026 Atmos Chem Phys*Sub-kilometre modelling of mineral dust emission from proglacial valleys*

A sub-km WRF-Chem implementation built to resolve dust emission from the proglacial valleys of the St. Elias Mountains, Canada — sources absent from current emission modelling and unaccounted for in global inventories. Land-surface inputs were rebuilt with explicit treatment of glaciofluvial deposit heterogeneity and inundation state; 5–19 d simulations across 2019–2022 were evaluated against in situ meteorology, dust measurements, camera stations and surface Doppler LiDAR. Total emission is estimated at $1.0 \times 10^4 \text{ kg km}^{-2} \text{ d}^{-1}$ from erodible deposits over 47 simulated days, offered explicitly as a first-order figure. Skill in reproducing surface meteorology and in-valley vertical dispersion is **season-dependent**, and the emission scheme remains sensitive to soil texture, so the flux estimate inherits that sensitivity. High-latitude dust is a genuinely under-measured PM source and this is a careful first quantification, but 47 days at one mountain range does not support extrapolation.

doi: 10.5194/acp-26-12613-2026

Taiwan LRT bioaerosol 2026 Environ Res*Long-range transport effects on airborne fungal concentration and persistence*

Daily wintertime fungal monitoring in northern Taiwan, comparing periods before, during and after Asian dust storms and frontal pollution events. Long-range transport raised fungal concentrations by **57–193% during events** and **17–69% in the post-event window**, with elevation persisting **one to four days** after passage. Frontal pollution enhanced concentrations more broadly than dust events did, and the assemblages differed by regime: dust storms enriched Basidiospores and *Cladosporium*, frontal events enriched *Aspergillus/Penicillium* and Ascospores. The persistence result is the useful one, because event-triggered sampling designs that stop at event end will systematically under-count exposure. Weaknesses are the usual ones for this design: event periods are defined meteorologically and confounded with the temperature, humidity and mixing changes that accompany them, and no source apportionment separates transported spores from locally resuspended ones.

doi: 10.1016/j.envres.2026.125640

Taiwan PM_{2.5}-bound metals 71 sites 2026 Res Sq (preprint)*PM_{2.5}-Bound Metals in a High-Tech Industrial Area of Northern Taiwan: Spatial Distribution, Sources, and Health Risks*

71 ambient sites sampled across two campaigns (2019, 2020) in a northern Taiwan high-tech industrial region. Mean PM_{2.5} was $19.33 \pm 8.97 \mu\text{g m}^{-3}$, higher in autumn (23.51) than summer (14.57), with hotspots near industrial parks; total metals averaged $736.85 \pm 751.20 \text{ ng/m}^3$, dominated by Zn, Ba, Cu, Pb and Mn. PMF gave fossil-fuel combustion 33.1%, soil and road dust 29.3%, metal industry 22.3% and tyre/brake wear 22.3%. Monte Carlo risk assessment returned a combined lifetime incremental cancer risk of 2.60×10^{-4} , driven by As and Cr, and a children's hazard index of 4.65 against 0.69 in adults, mostly via dust ingestion. A preprint, not peer-reviewed; source percentages sum past 100% as printed; two short campaigns cannot support a lifetime risk integral. The usable claim is that PM_{2.5} mass alone misranks risk where composition varies - the case for composition-aware monitoring.

doi: 10.21203/rs.3.rs-10850297/v1

Vilnius grey-green PM_{2.5} SHAP 2026 J Environ Manage*Unraveling associations between grey-green spatial trade-offs and PM_{2.5} concentrations using explainable machine learning*

Vilnius elderships 2016–2024 under a Grey-Green Space Coordination Index with LISA clustering and XGBoost-SHAP. GGSCI is negatively associated with PM_{2.5}; persistent high-high PM_{2.5} clusters sit on imbalanced grey-green structures. SHAP flips sign by range: NDVI above ~0.3 contributes negatively in balanced classes, while population density above ~2,000 km² and nighttime light above ~18 contribute positively in imbalanced ones. The awkward result, stated but not resolved, is that the most imbalanced GGSCI classes show the lower pollution - which inverts the paper's own framing and points to confounding by peri-urban land use. SHAP attributions are model explanations, not causal effects, and the PM_{2.5} field is itself modelled at a resolution the index outruns.

doi: 10.1016/j.jenvman.2026.130895

Xiamen chlorine NO₂ uptake 2026 Atmos Chem Phys*Chlorine-enhanced nocturnal NO₂ uptake and its nitrate yield in a coastal city*

Field observations in coastal Xiamen combined with a machine-learning attribution and a multiphase box model, contrasting sea-land-breeze nights against non-SLB nights. HONO and particulate nitrate both rose on SLB nights, with a mean NO₂ uptake rate constant of $k_{\text{NO}_2} = 9.70 \times 10^{-6} \text{ s}^{-1}$; chlorine ranked as the single most important driver of the enhancement. Feeding that constant back into the box model resolved the standing underestimation of both HONO and nitrate production, with nocturnal uptake accounting for **83.9% of HONO** and **46.4% of nitrate formation**. The machine-learning importance ranking is associational and cannot separate chlorine from the marine air mass that carries it, and one coastal city in one season limits transfer. Relevance is to secondary inorganic PM: nitrate is a large and poorly forecast fraction of coastal urban PM_{2.5}.

doi: 10.5194/acp-26-12715-2026

Cardiovascular & metabolic**8 records**

CARRIED FROM THE DAILY ISSUES — FULL SUMMARIES (8 OF 8 RECORDS IN THIS SUBTOPIC)

CHARLS coexposure 2026 Int Arch Occup Environ Health*Indoor solid fuels, outdoor PM and heatwave vs. cardiometabolic multimorbidity*

Five CHARLS waves 2011–2020, four disease cohorts (CMM $n = 10,273$; diabetes 9,893; heart disease 9,536; stroke 10,466), time-varying Cox plus a cumulative risk index and quantile g-computation. Single-exposure: solid fuel **HR 1.118 (1.016–1.231)** for heart disease and **1.149 (1.007–1.310)** for stroke; PM₁, PM_{2.5} and PM₁₀ were the most consistent risk factors ($P < 0.001$). Mixture ORs: **CMM 1.650 (1.375–1.979)**, **diabetes 1.675 (1.475–1.903)**, with solid fuel carrying the largest weight. Fuel use is self-reported and correlates with income, education and rurality, so the weight assigned to it is the estimate most exposed to residual confounding. Outdoor PM is modelled, not measured, at residence.

doi: 10.1007/s00420-026-02232-4

California PM_{2.5} diabetes mortality 2026 Front Public Health*Associations of PM_{2.5} exposure with diabetes-related mortality for California residents*

60,824 diabetes-related deaths (ICD-10 E11, primary or contributory) and 119,053 matched controls from California vital records, 2010–2021. Per 2.65 $\mu\text{g m}^{-3}$ IQR increase in the one-year rolling mean PM_{2.5}, conditional logistic regression gave OR 1.18 (95% CI 1.15–1.22), rising to 1.21 (1.17–1.25) after NO₂ adjustment - the association strengthening on co-pollutant adjustment is itself a signal that NO₂ is acting as a negative confounder here. The design's weakness is structural: controls are drawn from the same mortality file and are people who had not yet died, so both arms are decedents and the contrast is timing of death, not death versus survival. The \$31.2 million per 100,000 avoided-loss figure rests on that design.

doi: 10.3389/fpubh.2026.1903823

China WBC homeostasis 2026 Environ Pollut*Long-term exposure to ambient air pollution and circulating white blood cell homeostasis among adults in China: Time-window-specific associations and the mediating role of fasting blood glucose*

29,603 adults with routine health examinations at Liangshan General Hospital, 2016–2024, pollutants assigned to geocoded residences. Associations were strongest at the six-month window and were negative for every particulate pollutant: per IQR, WBC fell 0.503 $\times 10^9/\text{L}$ (95% CI 0.454–0.552) for PM_{2.5} and 0.349 (0.307–0.391) for PM₁₀, while O₃ was positively associated. Quantile g-computation put PM in the dominant negative position at six months and SO₂ at two years; fasting glucose mediated 4.97–10.98%. The direction is the problem: a fall in WBC with rising PM runs against the inflammatory-recruitment model the authors invoke, and a single-hospital examinee population selected on health-seeking is a plausible source of that inversion. Not included in the forest plot - the estimates are absolute counts, not ratios

doi: 10.1016/j.envpol.2026.129140

Constituent multiomics cardiopulmonary 2026 Environ Sci Technol*PM_{2.5} Constituents and Cardiopulmonary Dysfunctions in Older Adults: A Multiomics Approach Unveiling Molecular Pathways*

Longitudinal panel, 313 repeated measurements in older adults, personal rather than ambient PM_{2.5} with constituent speciation, screened by a linear mixed model plus three machine-learning methods. Chloride, nickel, barium and selenium emerged as the key constituents. Per IQR, chloride was associated with a 3.48% (95% CI 1.50–5.46) rise in diastolic blood pressure and nickel with a 0.09% (0.05–0.12) fall in FEV1/FVC. Mediation implicated 212 CpG sites acting on both cardiovascular and pulmonary endpoints and 7 genes spanning more than two constituent-function pairs, with 22 CpGs linked by eQTM to 14 coding genes including CERS6 and LRCH3. Limitations: 313 measurements is a small panel for a screen over this many constituents and CpGs, the constituent set is collinear so the four winners are correlates rather than isolated causes, and the mediation is cross-sectional within visit. Personal monitoring is the strength - it removes the spatial misalignment that defeats constituent studies built on central-site speciation

doi: 10.1021/acs.est.6c04206

Hangzhou AAD PM components 2026 J Am Heart Assoc*Which PM_{2.5} component triggers acute aortic dissection, and by how much*

Time-stratified case-crossover on Hangzhou acute aortic dissections, 2015–2023, with conditional logistic regression for single components and weighted quantile sum for the mixture. At lag 0–1 days, interquartile-range increases in black carbon, organic matter and sulfate were associated with 42% (OR 1.42, 1.15–1.75), 35% (1.35, 1.10–1.64) and 36% (1.36, 1.09–1.69) higher odds of Stanford type A dissection; type B associations were positive but not significant. In the joint model **black carbon carried more than half the weight**. Age, smoking, alcohol use and hypertension modified the associations. The case-crossover design self-controls for time-invariant confounding, which is its strength; the weaknesses are that component concentrations are strongly collinear (WQS mitigates but does not resolve this), exposure is assigned from ambient monitors rather than personally, and a single city's aerosol composition determines which component appears to dominate.

doi: 10.1161/jaha.126.049278

Sichuan Basin CAA 2026 Int Arch Occup Environ Health*Temporal changes in short-term PM_{2.5}-cardiorespiratory mortality associations under the Clean Air Action*

Multi-county time-series, Sichuan Basin 2014–2022, extended two-stage design with county-specific estimates pooled and longitudinal meta-regression on annual excess risk. ER per 10 $\mu\text{g}/\text{m}^3$ traced 0.7% (0.6–0.9) → 0.4% (0.3–0.6) → 1.4% (1.2–1.6) across the three policy periods, $P < 0.001$ for the pattern; scenario decomposition gives 52,831 deaths averted by concentration decline against 10,041 added by the risk change, and the late rise held in both sexes and was larger at ages 65+. The threat to the inference is that the final period is 2020–2022: pandemic-era changes in care-seeking, mortality coding and time-activity are not separable from a true slope change in this design. Still the most consequential result here — a falling mean does not imply a falling per-unit risk, and networks must resolve smaller contrasts as policy succeeds.

doi: 10.1007/s00420-026-02237-z

UK PH metabolomic mediation 2026 Ecotoxicol Environ Saf*Air pollution, pulmonary hypertension, and how much of it runs through metabolism*

Prospective cohort of 444,346 participants free of pulmonary hypertension at baseline with NMR metabolomics, 2,328 incident cases over a 13.58-year median follow-up. PM_{2.5}, PM₁₀, NO₂ and NO_x entered individually and as a joint air pollution score. The score gave HR 1.12 per SD (95% CI 1.08–1.17); individual pollutants ran 1.06–1.93 per 10 $\mu\text{g}/\text{m}^3$. Elastic-net regression produced a 105-metabolite signature dominated by lipoprotein measures, fatty acids and amino acids, itself associated with incident PH (HRs 1.12–1.24 per SD), **mediating 11.74% (7.91–18.95%)** of the score–PH association. The mediation estimate is the part to hold loosely: metabolites were measured once at baseline, so the mediator is cross-sectional with respect to a 13-year outcome, and the causal ordering of exposure, metabolome and disease cannot be established from that design. Residential exposure models add the usual non-differential error, which biases the 11.74% downward.

doi: 10.1016/j.ecoenv.2026.120778

Zhongshan AMI 2026 Front Public Health*Daily changes in PM_{2.5} and acute myocardial infarction hospitalisation*

Individual-level time-stratified case-crossover, 1,795 AMI admissions to one tertiary hospital in Zhongshan 2020–2024, 5,877 control days matched on weekday within calendar month; residential PM_{2.5} from the CHAP 1 km product; conditional logistic regression adjusted for temperature, humidity and holidays. A 10 $\mu\text{g}/\text{m}^3$ lag0–1 increment gave +7.89% (0.86–15.40) odds, with 22.58% in women, 22.16% at ages >65, 10.85% in the cold season and 17.90% for STEMI; age modified the effect ($P_{\text{int}} < 0.05$). Robust to co-pollutants except NO₂, which is the honest weakness — traffic co-exposure is not excluded. Single-hospital catchment limits generalisability, and the 1 km modelled field is assigned to address, not measured. The transferable idea for sensing is the first-difference metric: it cancels the additive bias that dominates low-cost PM error.

doi: 10.3389/fpubh.2026.1911027

4HAIE hair cortisol 2026 Stress

Intensity-dependent associations of physical activity and cardiorespiratory fitness with hair cortisol concentrations in the context of lifetime air pollution exposure

133 adults (87% women) from the 4HAIE industrial-environment study. Hair cortisol over ~3 months by LC-MS/MS against Fitbit and questionnaire activity, VO₂peak, and lifetime PM_{2.5}/PM₁₀/NO₂ reconstructed from residential histories. Vigorous activity was positively associated with hair cortisol (LTEQ B = 0.010, p = 0.042; Fitbit B = 0.006, p = 0.044); VO₂peak attenuated to null once vigorous activity entered. Lifetime air pollution showed no association with hair cortisol. A clean null, but an underpowered one: n = 133 against a lifetime exposure contrast within one industrial region leaves little variance to detect, so this constrains nothing about the PM-cortisol question

doi: 10.1080/10253890.2026.2721526

ABCD amygdala 2026 (preprint) bioRxiv (preprint)

Air pollution, early-life adversity and amygdala volume in preadolescents

3,601 ABCD Study children aged 9–10, annual residential PM_{2.5} total mass plus 15 components, linear mixed-effects adjusted for sociodemographics, co-pollutants and imaging covariates. Adversity predicted general, internalising and externalising symptoms; **PM_{2.5} moderated the adversity-externalising association**, with the highest symptoms in children high on both, and smaller basolateral paralaminar volumes tracked symptoms within the high-adversity group. Cross-sectional, so the mediation reading the framing invites is not supported; annual average exposure at residence cannot speak to the timing the developmental argument needs; and with 16 exposure terms the component results want a correction the abstract does not describe. Not peer reviewed. The component panel is the part worth watching — it is the kind of speciated contrast that mass-only networks cannot support.

doi: 10.64898/2026.08.31.748438

ENVIRONAGE eye-track 2026 EBioMedicine

Early-life air pollution and oculomotor markers of cognitive workload

196 children aged 9–11 from the ENVIRONAGE birth cohort completed eye-tracking paradigms 2021–2023; residential BC, NO₂ and PM_{2.5} from a high-resolution spatiotemporal model combining fixed-site monitors, land use and dispersion; multiple linear regression then BKMR distributed-lag models across windows. After multiple-testing correction the surviving associations are *recent* exposure with saccadic outcomes — per IQR of last-month BC **−10.97°/s (−18.67, −3.26)** and NO₂ **−10.81 (−18.13, −3.49)** saccade velocity — plus first-trimester and first-1000-days windows on the same outcome. Pupil diameter, fixation count and attention-bias outcomes did not survive. n = 196 against many windows and pollutants means the surviving hits are the ones to trust and the null ones are underpowered rather than negative. Notable for sensing: the effect attaches to last-month exposure, a window a personal or neighbourhood monitor can actually measure.

doi: 10.1016/j.ebiom.2026.106469

Older-adult cognition review 2026 Rev Environ Health

Air pollution and cognitive outcomes in older adults: a systematic review

PubMed searched to February 2025 for observational studies from 2000 onward on ambient air pollution and cognition in adults 50+; 36 studies included. PM_{2.5} is the most-studied pollutant, and most studies report worse global or domain-specific cognition, faster decline, or higher impairment and dementia risk with higher PM_{2.5}. Other pollutants trend adverse; ozone is the least consistent. Heterogeneity is attributed to exposure-assessment method, cognitive instrument, design and confounding control. Two limitations bound what this can support: a single-database search with no quantitative pooling and no formal risk-of-bias tool, so the conclusion is a vote count, not a weighted synthesis, and publication bias in a literature this size is unaddressed. The exposure-assessment heterogeneity it names is the actionable part

doi: 10.1515/reveh-2026-0022

Passive smoking acute stress 2026 Front Public Health

Self-reported secondhand smoke and acute stress in 4,460 rescue personnel

Cross-sectional survey of 4,460 grassroots rescue officers and soldiers, screening with the Acute Stress Disorder scale and the PCL-5. Univariate odds were **2.509 (1.621–3.883)** for acute stress disorder and **3.298 (1.459–7.451)** for PTSD; on full adjustment the ASD association survived and **the PTSD association did not (2.513, 0.900–5.151)**. Among never-smokers, exclusive passive exposure still carried OR 1.910 (1.071–3.408). Three problems limit what can be drawn from this. Exposure is self-reported with *no particle measurement of any kind*, so the dose is unknown and likely differentially misreported by people who are already distressed. The design cannot order exposure and outcome. And the reported multivariate ASD interval — 1.959–3.144 around a point estimate of 1.990 — is internally inconsistent, so the univariate estimate is the one plotted here. Filed as a signal that SHS exposure belongs in trauma screening, not as a dose-response result.

doi: 10.3389/fpubh.2026.1790527

Sex modification dementia meta 2026 Environ Pollut

126 million participants: sex does not modify the PM_{2.5}–Alzheimer's association

Systematic review and meta-analysis of 24 longitudinal cohorts, **over 126 million participants**, searched to September 2025, using the female-to-male ratio of risk ratios as the modification metric — the correct estimand, and one that many sex-difference papers substitute a pair of stratified estimates for. For Alzheimer's disease the pooled PM_{2.5} **RRR is 0.963 (95% CI 0.919–1.009)** and for vascular dementia **1.003 (0.954–1.054)**: null, and precisely so. The one positive is all-cause dementia (1.046, 1.024–1.069), a 4.6% relative excess in women that the authors decline to build policy on. Heterogeneity was negligible to low except PM₁₀ in Alzheimer's (*I*² = 61%), no publication bias was detected, and GRADE certainty is moderate. The stated limitation is the one that matters: methodological heterogeneity in how the constituent cohorts defined and adjusted sex could mask a true difference, so this is evidence against current *claims*, not proof of no effect.

doi: 10.1016/j.envpol.2026.129138

WTC responder exposure mixture 2026 Nat Commun

Urban exposures, frailty, and mental illness in World Trade Center Health Program responders

18,734 WTC Health Program general responders, later-life residential exposures. A decile increment in the joint PM_{2.5}/temperature/greenness mixture raises frailty index by 0.56% (95% CI 0.14–0.98) and PTSD score by 0.31% (0.07–0.55); effects are larger in responders with higher original WTC exposure, read as acquired biological vulnerability. The estimates are percentage differences on continuous scores, not ratio metrics, so they are small in absolute terms and the mixture decile has no physical unit - a decile shift is not a concentration. Residential assignment at annual resolution cannot separate PM_{2.5} from the temperature and greenness terms it co-varies with. For distributed sensing the relevant point is that the mixture framing needs co-located PM, thermal and greenness data at a resolution annual products cannot give.

doi: 10.1038/s41467-026-77401-7

Amazon PM_{2.5} x heat 2026 Research Square*Interaction Effects of PM_{2.5} and High Temperatures on the Respiratory Health of Vulnerable Age Groups in the Brazilian Amazon*

PREPRINT. Monthly municipal data 2009-2019 for the Legal Amazon, hierarchical Bayesian models with 0-2 month distributed lags. Per 10 $\mu\text{g m}^{-3}$ PM_{2.5} under high-temperature conditions, respiratory mortality in adults 60+ rose 2.5% and hospitalisations in those aged 1-19 rose 2.7%, both at lag 0 with no reported persistence at longer lags. Limitations: ecological design at monthly resolution, so the interaction is estimated well below the timescale on which wildfire smoke and heat actually covary, and no confidence intervals are given in the abstract. Relevance: a monthly aggregate cannot resolve smoke episodes, which is exactly the gap a dense low-cost network in the burning arc would fill

doi: 10.21203/rs.3.rs-10973377/v1

Bengaluru COPD 2026 Indian J Community Med*Prevalence and risk factors of COPD in an urban poor locality*

Community-based cross-sectional survey, adults over 30 in an urban poor locality of Bengaluru, PPS sampling to a calculated 1,560, spirometry-confirmed diagnosis. Prevalence 10.38% overall, 14.4% in men and 7.1% in women, with 74.06% at moderate-to-severe stage; multivariate logistic regression retained age, smoked and smokeless tobacco, fuel type, biomass exposure, environmental tobacco smoke and indoor air pollution. No air measurement of any kind — every exposure is a questionnaire item, so the “indoor air pollution” association is a self-report contrast and the effect sizes are not reported in the abstract. Value here is the spirometry-confirmed prevalence in a population that ambient monitoring networks systematically under-serve.

doi: 10.4103/ijcm.ijcm_145_25

COPD anaemia metals 2026 Int J Chron Obstruct Pulmon Dis*Air-pollution-associated intracellular metals, immune gene expression and erythrocyte indices*

61 COPD patients and 10 controls; annual PM₁₀, PM_{2.5}, NO₂ and NO_x by land use regression; metals by ICP-MS and single-cell ICP-MS in PBMCs and red cells; PBMC RNA-seq. PM_{2.5} was associated with higher RBC count ($\beta = 0.0314 \times 10^6 / \mu\text{L}$, 0.0066–0.0561), red cell distribution width, and with intracellular chromium, zinc and cadmium; those metals in turn tracked *ALPL*, *CXCR1* and *PI3* expression, and anaemic patients showed a distinct signature. Ten controls against 61 cases and a cross-sectional design make every link correlational, and LUR annual means cannot support a claim about cellular metal kinetics. Read it as an assay-development result: single-cell ICP-MS as an internal-dose readout is the transferable piece, and it is the kind of endpoint a speciating monitor could eventually be validated against.

doi: 10.2147/COPD.S626348

Cardiorespiratory review 2026 Med Gas Res*Narrative review of air pollution and cardiorespiratory disease*

Orientation-level review of the mechanisms linking ambient air pollution to cardiometabolic risk and cardiovascular death, with sections on heart failure, COPD and asthma exacerbation, on vulnerable groups (pregnancy, neonates, children, pre-existing cardiorespiratory disease), on the COVID-19 lockdown natural experiment, on international divergence in air quality standards, and on mitigation from policy through technology. There is no systematic search protocol, no PROSPERO registration, no pooled estimate and no risk-of-bias assessment, so it cannot be cited as evidence for any magnitude — and it does not claim to be. Listed because the lockdown-period material and the standards comparison are a useful map of where the regulatory targets currently sit, and because naming it here prevents it being mistaken for a systematic review later.

doi: 10.4103/mgr.MEDGASRES-D-25-00201

Disaster respiratory review 2026 Thorac Res Pract*Respiratory emergencies in disasters*

Narrative review spanning earthquake silica and asbestos dust, wildfire fine particulate, flood-related mould, volcanic ash and acidic gases, chemical release, blast barotrauma and pandemic threats, synthesised from PubMed/MEDLINE, Web of Science and Scopus with EM-DAT, WHO and UNDRR reports. It reports that acute respiratory infection is a leading cause of disaster-related morbidity and mortality across age groups, that chronic respiratory patients face elevated exacerbation risk, and that ARDS management remains hardest in resource-limited settings. There is **no stated search protocol, no inclusion criteria and no risk-of-bias appraisal**, so it indexes the literature rather than weighing it, and no quantitative estimate here can be traced to a defined evidence base. Carried as a citation anchor for the disaster-PM framing and for its explicit treatment of climate change as a driver of rising burden — not as evidence.

doi: 10.4274/ThoracResPract.2026.2026-6-13

Korea 8.25M asthma cohort 2026 Allergy*Cumulative Vulnerability to Particulate Matter and Incident Asthma: Joint Effects of Urbanicity, Poverty, and Age in a Nationwide Cohort*

8,252,498 adults in the Korean National Health Insurance Service database linked to 1-km long-term PM estimates, Cox models for incident asthma. Per 10 $\mu\text{g m}^{-3}$, PM_{2.5} gave HR 1.33 (95% CI 1.32-1.34) and PM₁₀ HR 1.38 (1.37-1.38). Stratification by age, Medical Aid status and urbanicity produced a monotone gradient from HR 1.16 (1.14-1.18) in younger rural adults without Medical Aid to 1.80 (1.66-1.94) in older urban Medical Aid recipients, p for interaction < 0.001. Limitation: incident asthma in claims data is diagnosis-coded, and urban Medical Aid recipients have higher healthcare contact, so part of the 1.16-to-1.80 gradient is plausibly ascertainment rather than susceptibility - the confidence intervals are narrow enough to make that distinction matter. Exposure contrast at 1 km is coarse relative to the urbanicity modifier being tested, which is the resolution argument for dense low-cost networks

doi: 10.1111/all.70502

Korea paediatric severity IV 2026 medRxiv*Air pollution drives concentrated clinical severity of respiratory illness in early childhood*

PREPRINT. Nationwide claims for 3,210,381 children, 99,261,945 child-months, 2015-2019, with thermal inversions used as a quasi-random instrument for monthly PM_{2.5}. A 1-SD rise ($7.23 \mu\text{g m}^{-3}$) adds 0.29 respiratory visits per child-month at age 1, significant through age 6; expenditure rises 46.7%, 43.0% and 35.4% at ages 0, 1 and 2 and is indistinguishable from zero past age 4, giving a 32.2% increase in annual respiratory spending under age 5. The severity gradient, not the visit count, is the contribution. Limitation: the inversion instrument is plausibly exogenous to care-seeking but not to temperature, and no temperature control is described in the abstract; that is the exclusion restriction the peer review will test

doi: 10.64898/2026.09.03.26362213

Kuwait tobacco smoke morbidity 2026 BMC Health Serv Res*Exposure to tobacco smoke, concurrent short-term symptoms of respiratory, allergic, cardiovascular morbidities and health-services use among young adults in Kuwait*

Convenience sample of 1,323 Kuwait University undergraduates, October 2022, electronic questionnaire. First-hand smoke exposure 13.4%, second-hand at home 50.8%, any exposure 55.9%. Self-reported symptoms were common: shortness of breath 59.1%, cough 46.3%, chest pain 26.9%. Limitations are severe and the authors name most of them: convenience sampling in one university, 75.8% female, all exposure and outcome self-reported, cross-sectional so no temporality. The prevalence figures are usable as a local exposure baseline; the symptom associations are not evidence of effect

doi: 10.1186/s12913-026-15530-x

MobiliSense ambulatory spirometry 2026 Research Square*Sub-hourly personal black carbon and a measurable drop in FEV₁*

Preprint. 199 adults in metropolitan Paris, 2018–2020, each carrying two portable pollutant monitors, a GPS receiver and an accelerometer through ordinary daily mobility, with **2,504 spirometry tests** taken morning and evening over three days. Mixed-effects models tested lags from 15 minutes to 6 hours. In multipollutant models, a 1 µg/m³ increase in BC one hour before testing was associated with FEV₁ lower by **0.016 L** (95% CI –0.024 to –0.008), and at two hours by **0.021 L** (–0.034 to –0.007); PM_{2.5} at 15, 30 and 60 minutes reduced FEV₁/FVC by 0.60, 0.70 and 0.50 percentage points. O₃ ran positive, which is the expected artefact of O₃ anticorrelating with traffic. The threats are real: 199 participants, three days, no personal-monitor validation reported against reference instruments, and spirometry quality in the field is hard. But this is the design that resolves an exposure window a fixed monitor cannot see at all.

doi: 10.21203/rs.3.rs-10386483/v1

PM_{2.5} lung cancer never-smokers review 2026 J Thorac Dis*Breathing carcinogens: PM_{2.5} air pollution and lung cancer in never-smokers, a narrative review*

Narrative review of PM_{2.5} and lung cancer in never-smokers, PubMed/MEDLINE 2000 to May 2026 plus reference-list snowballing. The central claim is mechanistic rather than quantitative: PM_{2.5} acts predominantly as a tumour promoter rather than a direct mutagen, operating through inflammatory signalling, metabolic reprogramming and epigenetic change, and is associated with more advanced stage at diagnosis and worse survival after resection. Lung cancer in never-smokers is described as molecularly and immunologically distinct, with screening and precision-oncology implications. No pooled estimate, no risk-of-bias assessment and no search-yield count are reported, so it carries no quantitative weight and the selection is author-curated by design. Useful as a mechanism map, not as evidence.

doi: 10.21037/jtd-2026-1264

Shijiazhuang AECOPD 2026 COPD*Seasonal and short-term environmental associations with COPD exacerbation admissions*

City-wide retrospective time-series of **22,468 AECOPD admissions**, 2020–2024, from the municipal medical insurance database covering all tertiary, secondary and community hospitals — unusually complete ascertainment, though ED discharges without admission are excluded. Mean age 71.7 (SD 9.1), male-to-female 3.7:1, winter peak 6,009 against summer 5,051. On the 7-day mean of lags 0–6, each 10 µg/m³ of winter PM₁₀ gave **IRR 1.069** (1.052–1.085) and NO₂ **1.089** (1.050–1.129); ozone was positive in summer **1.035** (1.015–1.056) and autumn **1.022** (1.005–1.039); a 1°C wider diurnal range gave **1.055** (1.034–1.077). The authors are explicit that the many lag × pollutant × season tests are exploratory and that the prediction models are not a deployable early-warning system — a welcome discipline, and the reason the apparent protective multipollutant IRRs should not be read as findings. Collinear pollutants in a single city cannot be separated by adjustment.

doi: 10.1080/15412555.2026.2723102

Umea wood smoke chamber 2026 Environ Res*Controlled wood smoke exposure, sensory irritation and small airway mechanics*

Randomised, double-blind crossover chamber study: **14 healthy non-smokers** (11 M, 3 F, mean age 26) exposed for 2 h to wood smoke at **PM₁ 453 ± 13 µg/m³** and to filtered air, three-week washout, with forced oscillometry (tremoFlo) alongside spirometry and Borg symptom scoring. Wood smoke produced significant **nasal irritation** ($p = 0.003$) with trends for cough and odour, but **no exposure-specific change** in any spirometric or oscillometric resistance parameter (R5, R5–R19, R5–R20). Reactance and resonant frequency moved with time irrespective of exposure, which is the control the design exists to provide. With $n = 14$ healthy young adults this is badly underpowered for small-airway effects and the null should be read as such, not as evidence of no effect — the authors say as much. The methodological point stands independently: irritative symptoms preceded any mechanical signal, so spirometry-only designs will under-detect early response.

doi: 10.1016/j.envres.2026.125629

Reproductive & developmental**6 records**

CARRIED FROM THE DAILY ISSUES — FULL SUMMARIES (6 OF 6 RECORDS IN THIS SUBTOPIC)

ECHO prenatal wildfire PM_{2.5} 2026 Environ Epidemiol*Prenatal exposure to wildfire fine particulate matter and fetal growth outcomes in the United States: Findings from the ECHO Cohort*

20,034 births (2006–2020) across 30 ECHO sites with daily census-tract wildfire PM_{2.5} and smoke-day counts. Mean pregnancy wildfire PM_{2.5} and smoke days were not associated with term birth weight, birth-weight-for-gestational-age z-score or small-for-gestational-age, and distributed-lag and trimester models showed no consistent sensitive window. The single positive signal is large-for-gestational-age, OR 1.015 per smoke day (95% CI 1.001–1.029), which is borderline and biologically unexplained. This is a well-powered null and should be read as one rather than as a failure; the binding limitation is tract-level assignment of a plume-scale exposure, which attenuates any true effect toward the null. Exactly the misclassification that dense low-cost networks exist to reduce.

doi: 10.1097/ee9.0000000000000526

France SUDI case-crossover 2026 Paediatr Perinat Epidemiol*Short-term air pollution and sudden unexpected death in infancy, national registry*

1,078 SUDI cases from the French national registry, 2015–2022, with modelled daily PM₁₀, PM_{2.5}, NO₂ and O₃ assigned at each infant's residence. A time-stratified case-crossover design with conditional logistic regression, adjusted for public holidays, influenza-like illness and daily mean temperature, examined lags to six days. **PM exposure was consistently associated with higher odds of SUDI at lags 3–5 and across the last week of life**, strongest in cumulative and multi-pollutant models; NO₂ showed a trend only, and effects were larger in spring. The case-crossover design self-matches on fixed confounders, which is its strength, but the abstract reports no point estimates or intervals — so the association's magnitude cannot be judged and this record contributes nothing to the forest plot. Residence-level modelled exposure also ignores where infants actually sleep and travel. SUDI remains an aetiologically open outcome; a short-term environmental trigger is a plausible and previously little-tested component.

doi: 10.1111/ppe.70196

Korean birth-cohort growth mixture 2026 Environ Pollut*Prenatal bisphenols and ambient air pollutants are differentially associated with early childhood length and weight from birth to 36 months: a multi-domain mixture analysis in a Korean birth cohort*

METADATA ONLY. No abstract was available on the entry date from Crossref per-DOI, Europe PMC, PubMed efetch or Semantic Scholar. Title establishes a Korean birth cohort with a multi-domain mixture analysis placing prenatal bisphenols alongside ambient air pollutants against length and weight trajectories to 36 months, and states that the two exposure domains behave differently - but not in which direction, for which pollutant, or at what magnitude. Nothing is asserted here and the record is tier C. Flag for re-reading: multi-domain mixture methods that carry chemical and air exposures in one model are the methodological question this archive should track

doi: 10.1016/j.envpol.2026.129145

Preeclampsia nationwide 2018-23 2026 Eur J Intern Med

METADATA ONLY — nationwide air pollution and preeclampsia, 2018–2023

The PubMed record carries no abstract, and neither Crossref nor Europe PMC held one on the entry date. What the record establishes: a nationwide study covering 2018–2023 has examined the association between ambient air pollution and preeclampsia. What it does not establish: the country, the pollutant set, whether PM was resolved at all, the case count, the exposure-assignment method, the trimester windows examined, or the direction and magnitude of any association — none of which follows from the title, and none of which is asserted here. Flagged for retrieval. Hypertensive disorders of pregnancy are one of the better-powered PM endpoints and a nationwide six-year series would be a substantial addition if the exposure model holds up.

doi: 10.1016/j.ejim.2026.107175

Prenatal greenspace ASD window 2026 Environ PollutA mid-to-late pregnancy window for green space — and PM_{2.5} is not the mechanism

1:3 age- and sex-matched case-control from a tertiary maternal and child health hospital in central China: 207 children with autism spectrum disorder, 621 controls. Monthly residential NDVI and PM_{2.5} were estimated from 10 months before to 3 months after birth and entered a distributed lag nonlinear model, with product terms testing NDVI × PM_{2.5} interaction, bootstrapped mediation, and Bonferroni correction. Cases had lower NDVI in all 14 months; the protective window localises to mid-to-late pregnancy, roughly 6 to 1 month before term birth. The negative result is the one that earns the entry: **PM_{2.5} neither mediated nor modified the green-space effect** after correction, which closes the most obvious alternative explanation rather than leaving it hanging. Caveats are the standard ones for this design — residential address as the exposure proxy, hospital-based case ascertainment, 207 cases against a 14-month lag structure, and green space as a proxy for an unmeasured bundle (noise, activity, socioeconomic position).

doi: 10.1016/j.envpol.2026.129118

SIP prenatal Bayley 2026 Child NeuropsycholIn utero PM_{2.5} and NO₂ and cognitive-behavioural development at 1–3 years

Stress in Pregnancy cohort, $n = 210$, Queens, New York; Bayley-III and ABAS-3 at 18, 24 and 36 months; prenatal PM_{2.5} and NO₂ taken from public monitoring databases. The reported pattern is **sex-specific and pollutant-specific**: PM_{2.5} associated with poorer communication, social skills and gross motor function in **boys**, NO₂ with poorer communication, gross motor and social-emotional skills in **girls**. No effect estimates or intervals appear in the abstract, which is why nothing from this paper enters the forest plot. With $n = 210$ split by sex across three timepoints and multiple Bayley and ABAS subdomains, the multiplicity burden is severe and the crossed sex-by-pollutant pattern is exactly the shape that subgroup analysis produces by chance. Exposure from public monitors rather than a spatial model adds classical misclassification. Hypothesis-generating, and the authors largely frame it that way.

doi: 10.1080/09297049.2026.2723475

Mechanistic toxicology**9 records**

CARRIED FROM THE DAILY ISSUES — FULL SUMMARIES (9 OF 9 RECORDS IN THIS SUBTOPIC)

6PPD-Q AC16 p53 2026 Ecotoxicol Environ Saf

Cardiotoxic mechanism of the tire-derived transformation product 6PPD-quinone

Network toxicology and docking followed by wet-lab confirmation in human AC16 cardiomyocytes. **144 putative targets** with TP53 the top hub; IC₅₀ **13.92 μM** at 24 h. 6PPD-Q drove ROS accumulation and **p53 Ser15 phosphorylation without raising total p53 protein or TP53 mRNA** — i.e. post-translational stress activation — alongside TRAF6 upregulation, MAPK1 phosphorylation, IL6/TNF induction, mitochondrial apoptosis (membrane potential loss, Bax/Bcl-2 rise, cleaved caspase-9 and -3) and blocked late autophagic flux (LC3-II accumulation with p62 depletion). Pifithrin-α attenuated all of it, which is the load-bearing experiment. The exposure gap is total: a micromolar dose to an immortalised cardiomyocyte line has no established relationship to any airborne 6PPD-Q concentration, and no inhalation route is demonstrated. Relevant here only because 6PPD-Q is a measurable constituent of road dust and traffic-derived PM.

doi: 10.1016/j.ecoenv.2026.120754

AS-IV vs PM vascular senescence 2026 Cardiovasc Toxicol

Replicative and Air Pollution-Induced Senescence: Telomere Dysfunction, Mitochondrial Stress, and Modulatory Effects of Astragaloside IV in Endothelial and Vascular Smooth Muscle Cells

HUVECs and human coronary smooth muscle cells under replicative senescence (passages 6-7, 10-11, 17-20) and fine-dust-induced premature senescence (48 h, 50 and 100 ug/mL certified urban PM₁₀-like reference material), then Astragaloside IV at 50 uM for 24 h. Both cell types showed SA-beta-gal positivity, telomere shortening, p16INK4a/p21Cip1 activation and mitochondrial dysfunction; fine dust drove premature senescence, telomere shortening, TERRA upregulation and acute mitochondrial impairment mainly in endothelial cells, with smooth-muscle cells relatively resistant. AS-IV partially rescued endothelial but not smooth-muscle damage. The dose is the problem: 50-100 ug/mL in culture is orders of magnitude above any inhaled ambient mass dose, so the mechanism is plausible and the magnitude is not transferable.

doi: 10.1007/s12012-026-10173-0

Berberine PM_{2.5} breast cancer 2026 ApoptosisBerberine shows potential in mitigating PM_{2.5}-induced breast cancer progression by inducing DNA damage and inhibiting error-prone DNA repair pathways

Water-soluble Delhi PM_{2.5} applied to MCF7 breast cancer cells increased proliferation, migration and ROS and produced multinucleated giant cells, i.e. genomic instability. Transcriptomic profiling gave a PM_{2.5} carcinogenic signature enriched in MAPK, ROS, metabolic, lysosomal and ribosomal pathways, with BIRC5, WSB1 and RCC1 dysregulated in patient tumours and reversed by berberine. The inference is weak on two counts: only the water-soluble fraction was used, which discards the insoluble carbonaceous and metal-bound mass that carries much of PM_{2.5} oxidative potential, and no dose-response or in vivo confirmation is reported. Relevant to sensing only as a reminder that mass concentration is a poor proxy for the bioactive fraction.

doi: 10.1007/s10495-026-02423-x

Fetal macrophage FCER1G 2026 Nat Commun*Prenatal PM_{2.5}, fetal macrophage epigenetic reprogramming and atopic dermatitis*

Birth-cohort association between **first-trimester** PM_{2.5} and early-childhood atopic dermatitis, carried into mechanism by integrative multi-omics including single-cell profiling of placenta and fetal immune cells. The central finding is **hypomethylation of FCER1G in fetal macrophages** driving sustained overexpression that persists across developmental stages and re-emerges in M2 macrophages of AD skin and peripheral blood; functional work links FCER1G networks to NADPH-oxidase-mediated ROS and Th2 inflammatory signalling. This is a rare case of a durable, cell-type-resolved epigenetic mediator proposed for a prenatal PM_{2.5} association rather than asserted. The exposure remains a modelled residential estimate assigned to a trimester, which is the weakest link in an otherwise strong chain — first-trimester windows are precisely where exposure misclassification is largest. The persistence claim rests on cross-sectional comparisons at different stages, not on following the same cells.

doi: 10.1038/s41467-026-76298-6

IL23A / NF-κB lung tumour 2026 Ecotoxical Environ Saf*IL23A acts as a pivotal mediator of PM_{2.5}-induced lung tumorigenesis through the NF-κB pathway activation*

PM_{2.5} raised proinflammatory cytokines including IL23A in lung epithelial and NSCLC lines; induction was blocked by p65 siRNA knockdown or chemical NF-κB inhibition, placing IL23A downstream of canonical NF-κB. IL23A is overexpressed in LUAD tissue and inversely associated with prognosis in non-smokers. Anti-IL23A neutralising antibody suppressed tumour formation and growth in orthotopic PM_{2.5}-treated mice. An atypical IL12B-independent secreted IL23A form was identified in lung epithelium. Limitation: PM_{2.5} dose and composition are not stated in the abstract, and instilled or extract-based dosing routinely exceeds inhaled ambient dose by orders of magnitude, so the axis is established and the exposure-response is not

doi: 10.1016/j.ecoenv.2026.120742

Nasal PM_{2.5} testicular pyroptosis 2026 Res Sq (preprint)*Nasal Exposure to Environmental PM_{2.5} Drives Testicular Pyroptosis and Tissue Damage: Evidence from Inhibitor Intervention and Multi-Source Data*

24 male Sprague-Dawley rats in four arms of six: saline, Caspase-1 inhibitor Z-YVAD-FMK, intranasal PM_{2.5} 10 mg/kg, and both, for four weeks. PM_{2.5} reduced sperm count and motility and raised abnormality rate, with elevated ASC, Caspase-1 and gasdermin D in testis, all reversed by the inhibitor - a coherent NLRP3/Caspase-1/GSDMD pyroptosis chain. Reanalysis of GEO GSE189187 (GC-2spd cells, 3 vs 3) gave 217 DEGs enriched in interferon/antiviral processes, with Hmox1 up and Stat1 down. n=6 per arm and a 3-vs-3 public dataset cannot support hub-gene claims, and intranasal instillation at 10 mg/kg bears no quantitative relation to ambient inhalation. Preprint, no peer review.

doi: 10.21203/rs.3.rs-8701799/v1

Retinal CRM 2026 Biochem Biophys Rep*Certified particulate reference materials in primary mouse retinal cells*

Two certified environmental particulate reference materials applied to primary mouse retinal cells, plus repeated ocular-surface exposure in mice. Effects on DCFDA fluorescence and ATP-dependent viability were condition-dependent and concentrated in glial conditions; neither α-tocopherol nor N-acetylcysteine gave a significant rescue, and glial coculture did not improve retinal ganglion cell survival. In postnatal mice topical particulate was associated with inner retinal thinning; in adult Thy1-CFP mice the within-animal comparison was null and only a *post hoc* between-animal pooling reached significance (**-14.4 cells, -28.1 to -0.7, P = 0.040**). The authors state plainly that this establishes neither particle transport to the retina nor a causal oxidative mechanism nor any ambient risk — an unusually disciplined set of negative caveats, and the reason this is tier C rather than a finding. Ocular-surface dosing is not inhalation, and certified reference material is not ambient PM.

doi: 10.1016/j.bbrep.2026.102758

Tire wear particle phytotoxicity 2026 Curr Biol*Tire wear particles stunt a roadside grass and raise root oxidative stress*

Controlled tire-wear-particle exposure of *Bromus catharticus*, a common roadside weed in Japan, testing an endpoint the TWP literature has largely skipped — previous work concentrated on aquatic systems and on crops, not on the roadside and urban-park vegetation that sits metres from the source. Both **shoot and root growth were significantly inhibited**, and root malondialdehyde — a lipid-peroxidation marker — rose significantly, indicating oxidative stress as at least one operative mechanism. The abstract does not give the applied dose, the particle size distribution, the exposure route (soil versus foliar) or the exposure duration, and all four determine whether the treatment is environmentally realistic; a single annual grass in a pot cannot stand for roadside vegetation generally. Carried here because TWP is the fastest-growing non-exhaust PM source, is poorly sized by optical monitors, and now has a documented terrestrial effect at the point of emission.

doi: 10.1016/j.cub.2026.06.087

Vancouver DE blood-lung concordance 2026 Environ Res*Concordant transcriptomics analysis of the lungs and blood after exposure to diesel exhaust: a randomized, controlled human exposure study*

Double-blind crossover controlled human exposure, 32 participants, two hours of diesel exhaust and of filtered air on separate occasions, with paired-end RNA-seq on blood and bronchial brushings. Airway DEGs at FDR < 10% include upregulated GPX2 and NQO1, i.e. the NRF2 antioxidant arm. Hallmark pathway mapping gives IFN-α, IFN-γ, inflammatory response and ROS activity in both compartments, and a concordant analysis finds correlated blood-airway expression that neither compartment showed alone. The authors are explicit that this is exploratory. n = 32 at a single two-hour dose, no exposure-response, and the concordance is correlational - but it is the right design question: whether a blood transcriptomic readout can stand in for bronchoscopy in panel studies of traffic-related particles

doi: 10.1016/j.envres.2026.125625

Occupational & indoor**15 records**

CARRIED FROM THE DAILY ISSUES — FULL SUMMARIES (15 OF 15 RECORDS IN THIS SUBTOPIC)

Auditorium IAQ + 16S bioaerosol 2026 Sci Total Environ*Indoor Air Quality (IAQ) analysis and 16S rRNA gene sequencing of indoor air pollutants in a 3000-capacity religious auditorium*

METADATA ONLY. Metadata only; abstract absent from Crossref, Europe PMC and PubMed on per-DOI retry. Retained because a 3,000-capacity assembly space is a high-occupancy indoor microenvironment where occupant-generated coarse particles and bioaerosol dominate, and 16S sequencing of airborne material is particulate characterisation. PM scope is inferred from the title, no finding asserted, and the study location is not recoverable from metadata. On the retry list.

doi: 10.1016/j.scitotenv.2026.182237

Bi/Bi₂Te₃ workplace ICP-OES 2026 Zhonghua Lao Dong Wei Sheng Zhi Ye Bing Za Zhi*ICP-OES determination of bismuth and bismuth telluride in workplace air*

Method development for an occupational aerosol with no established Chinese workplace standard: mixed cellulose ester filter sampling, ultra microwave digestion at 130°C for 10 minutes, then simultaneous Bi and Te by ICP-OES with Bi₂Te₃ back-calculated from either element using on-site speciation information. Linear over **0.04–30 mg/L** ($r > 0.9990$); on a 75 L sample diluted to 50 mL the minimum detectable Bi₂Te₃ is **0.006 mg/m³** via Bi and **0.008** via Te. Digestion efficiency 99.1–100.0% (Bi) and 101.2–103.3% (Te), intra-batch precision under 0.8%, 7-day room temperature stability, and no interference from Ag, Al or As up to 10 mg/L. Validated on simulated samples rather than a field inter-laboratory comparison, and thermoelectric workplaces are the only stated application. Included as the kind of low-visibility analytical groundwork on which occupational exposure limits eventually rest.

doi: 10.3760/cma.j.cn121094-20250319-00107

CFRP inhalation risk review 2026 Regul Toxicol Pharmacol*Carbon-fibre composite dust: fibre-like fragments, ultrafines, and thin evidence*

Life-cycle review of inhalation risk from carbon fibre-reinforced plastics across manufacturing, machining, recycling and end-of-life processing. Mechanical processing generates respirable high-aspect-ratio fragments; thermal processing generates ultrafine particles together with hazardous gases. The review's central judgement is a refusal: available evidence **does not demonstrate a magnitude or pattern of inhalation toxicity comparable to asbestos or to pathogenic high-aspect-ratio carbon nanotubes**, but it is too limited and heterogeneous — particularly for chronic exposure — to support any quantitative relative-toxicity comparison in either direction. It identifies the actual gaps: real-world occupational exposure data, harmonised emission scenarios, internal dose and biopersistence. As CFRP volumes rise with wind-blade and aerospace retirement, this is a source class whose workplace aerosol nobody is routinely monitoring.

doi: 10.1016/j.yrtph.2026.106243

China gym co-exposure 2026 Environ Pollut*Association of indoor multiple airborne pollutants co-exposure and typical respiratory symptoms among gym employees: A nationwide observational study in China*

280 gyms across 32 Chinese provinces, 2022-2023. Median PM_{2.5} 35 µg m⁻³, PM₁₀ 59 µg m⁻³, fungal count 170 CFU m⁻³, plate count 226 CFU m⁻³; 22.13% of employees reported respiratory symptoms. PM_{2.5} OR 2.31, PM₁₀ 2.19, fungi 1.17, bacteria 1.24, and a quantile g-computation mixture OR 1.70 with PM_{2.5} and fungal count as the dominant components. Natural ventilation (OR 0.36) and high-efficiency filters (0.42) were protective; absent equipment disinfection (3.30) and air-return air conditioning (2.85) raised risk. Limitations: cross-sectional, self-reported symptoms, and no confidence intervals reported for any OR in the abstract, which is why the forest plot carries them as point estimates only. The ventilation and filtration coefficients are the operationally useful part

doi: 10.1016/j.envpol.2026.129141

Dutch school air cleaner RCT 2026 Front Public Health*Protocol: 180 classrooms, HEPA versus ionisation versus nothing*

Protocol only, no results (ClinicalTrials.gov NCT07479420). 180 classrooms in 29 Dutch primary schools, clustered within schools and randomised to HEPA-filter portable air cleaners, ionisation/plasma cleaners, or none. Units were laboratory pre-tested, safety screened and operated at comparable clean air delivery rates — which is the design choice that makes the comparison mean something, since CADR is the confounder that sinks most field air-cleaner studies. The schedule is a three-week baseline or two-week post-intervention control plus three repeated three-week intervention periods, to 14 weeks per school. Airborne dust is collected passively on electrostatic dustfall collectors and assayed for bacterial and viral markers, with active sampling in 12 classrooms to validate the passive method; PM₁₀, PM₄, PM_{2.5}, PM₁, CO₂, temperature, humidity and VOCs are logged continuously. Absenteeism and parent-reported symptoms are retrospective, which is the weakest element. Worth tracking to results — ionisation devices rarely get a fair, CADR-matched test.

doi: 10.3389/fpubh.2026.1892902

Ethiopia dormitory ventilation 2026 Indoor Environments*METADATA ONLY — occupant-driven ventilation in naturally ventilated dormitories*

No abstract retrievable on the entry date. The title commits to a field study of occupant-driven ventilation and indoor exposure dynamics in naturally ventilated Ethiopian university dormitories. The pollutants measured, the ventilation-rate method (tracer gas, CO₂ decay or modelled), the number of rooms and the season are all unrecoverable and none is asserted here. Occupant-controlled natural ventilation is the largest uncontrolled term in indoor exposure models for warm low-resource settings, and measured air-change data from Sub-Saharan Africa are scarce enough that the study is worth retrieving on those grounds alone.

doi: 10.1016/j.indenv.2026.100192

Europe indoor PAH children 2026 Environ Res*Indoor-air PAH levels and child cancer risk across six European cities*

Twelve EPA PAHs in air and deposited dust from schools, homes and sports halls in Athens, Barcelona, Colchester, Copenhagen, Helsinki and Lisbon — four of the 16 target compounds were dropped for analytical limitations, which is stated. Spatial variability is large: medians above **50,000 pg/m³** where intensive residential wood burning and poor ventilation coincide, against **under 10,000 pg/m³** in south-western Europe. High-molecular-weight PAHs ran higher in schools than homes, consistent with occupancy and accumulation of outdoor-derived particles, while low-molecular-weight species tracked household practices. Estimating air concentrations from deposited dust **underestimated HMW-PAHs**, which is a directly useful negative result for anyone tempted to substitute dust for active sampling. Excess cancer risk over a five-year primary-school exposure period spanned **10⁻⁸ to 10⁻⁵**, below the 10⁻⁴ reference. Cross-sectional sampling cannot capture seasonal wood-burning cycles.

doi: 10.1016/j.envres.2026.125643

JECS indoor PM Kawasaki 2026 Int J Hyg Environ Health*Household-specific indoor PM_{2.5} and PM_{10-2.5} concentrations and Kawasaki disease onset by age six: the Japan Environment and Children's Study*

3,846 mother-child pairs in the JECS sub-cohort with two one-week household PM measurements. Cumulative Kawasaki disease incidence by age six was 15.34 per 1,000. Higher indoor PM_{10-2.5} was associated with consistently lower odds of Kawasaki disease across exposure categories, with a weaker inverse trend for PM_{2.5} - an inverse association the authors report without over-reading. Intraclass correlations of 0.52-0.55 between the two measurement occasions mean roughly half the variance in a household's exposure is not captured by these snapshots, and with ~59 expected cases the categorical ORs are thinly powered. Two one-week integrated samples as a six-year exposure surrogate is precisely the gap continuous indoor sensing would close.

doi: 10.1016/j.ijheh.2026.114893

Occupational nanoparticle scoping 2026 Part Fibre Toxicol*Occupational nanoparticle exposure and its effects on pulmonary health, cardiovascular health, and systemic biomarkers: a scoping review*

MEDLINE, EMBASE and Web of Science to April 2024; 66 of 6,020 records included, mostly cross-sectional ($n = 37$) or panel ($n = 18$). Exposures are dominated by carbon nanotubes and fibres, titanium dioxide and metal oxides in manufacturing. Reported outcomes are reduced lung function and raised inflammatory and oxidative-stress markers, with cardiovascular and systemic biomarker changes. The review explicitly performed no quality appraisal, so the evidence map is a catalogue of what has been measured rather than a weighing of it, and the cross-sectional predominance means reverse causation is unaddressed across most of the base. Useful mainly for the biomarker inventory and for locating where UFP metrology in workplaces is missing

doi: 10.1186/s12989-026-00699-0

Oklahoma cannabis facility IAQ 2026 Ann Work Expo Health*Seven-instrument occupational air-quality characterisation of a cannabis grow room*

Continuous monitoring from 10 July to 1 August 2023 in a 186 m² HVAC-equipped medical cannabis cultivation room holding roughly 900 plants on a 12:12 photoperiod. Seven instruments across three spatial zones at 1.5 m breathing height — three Aranet4 NDIR CO₂ sensors, two TSI AirAssure multi-parameter monitors carrying size-resolved PM, and two PurpleAir PA-II-FLEX dual-channel counters — produced ~113,000 quality-controlled records. CO₂ tracked the light cycle, with means of 979–1,086 ppm and maxima above 2,200; nighttime elevation is attributed to plant metabolism and enrichment rather than worker respiration, since workers leave at lights-out. **Median PM_{2.5} was low at 1.4–3.1 $\mu\text{g m}^{-3}$** with episodic activity-driven spikes. One facility, one room, one summer month, and no personal sampling, so this is a determinants pilot rather than an exposure assessment — but it is the rare occupational study that runs consumer-grade and research-grade particle instruments side by side.

doi: 10.1093/annweh/wxag074

PSH smoke-free home RCT 2026 JAMA Netw Open*A smoke-free-home trial that missed its primary outcome, reported plainly*

Cluster-randomised trial across 40 permanent-supportive-housing sites in the San Francisco Bay Area, 400 residents (mean age 54.5, 63% male) who smoked at home, January 2022 to March 2025, intervention sites receiving one-to-one coaching on adopting a smoke-free home and staff at all sites receiving brief cessation training. **The primary outcome failed:** 90-day smoke-free-home adoption was 6.8% versus 4.8%, **OR 1.45 (0.69–3.07)**. What moved was the softer secondary — any adoption of at least one day, 63.4% versus 36.8%, **adjusted OR 3.83 (2.63–5.57)** — and CO-verified 7-day abstinence, 6.3% versus 1.0%, OR 6.94 (1.69–28.45), which rests on 14 events in total and should not be read as an effect size. The honest reading is that brief coaching moves intention and not maintenance in a population with very high baseline exposure, and that a waiting-list control cannot rule out attention effects on the self-reported outcomes.

doi: 10.1001/jamanetworkopen.2026.32697

Rouyn-Noranda citizen dust 2026 Environ Monit Assess*Citizen-science measurement of smelter-derived metals in snow and indoor dust*

Residents of 23 homes 540–2,080 m from the Horne Smelter — Canada's only copper smelter — collected their own indoor dust and adjacent snow samples for Cu, Pb, Cd, As, Zn and Cr. Accounting for distance and wind direction, smelter location explained **73% of snow variance for Cu and Cr**, 58% for Cd and Pb, 56% for As and 44% for Zn ($p < 0.01$); for indoor dust the distance-only explanatory power was much lower at **36% (Cu) to 15% (As)**. Copper served as a smelter tracer and correlated strongly with the other metals in snow ($R^2 > 70\%$). Few residential characteristics predicted indoor dust once distance was controlled. Twenty-three homes is a small sample, snow and settled dust are deposition integrals rather than airborne concentrations, and participant-collected samples carry uncontrolled handling variance. The value is the demonstration itself: a participatory design produced a defensible source-attribution gradient in a community with a documented exposure dispute.

doi: 10.1007/s10661-026-15900-2

Tangshan CWP IGF1 2026 Front Public Health*Dust exposure, IGF1 and autophagy in coal workers' pneumoconiosis*

Retrospective analysis of 2,028 employees at one Tangshan coal enterprise (2023–2024) with a paired SiO₂-stimulated macrophage model. Logistic regression retained early initial working age, dust-exposure duration, smoking, white cell count and AST/ALT ratio; patients and stimulated macrophages both showed raised IGF1, altered LC3/P62 and increased PI3K/AKT/mTOR phosphorylation. Two weaknesses matter. Exposure is job history, not a measured respirable dust concentration, so no exposure–response can be estimated; and the cell model uses crystalline silica while the cohort's hazard is coal mine dust — these are different particles with different mechanisms, and the paper's own language stays correlative. The white-cell and liver-enzyme “risk factors” are as plausibly consequences of disease as causes.

doi: 10.3389/fpubh.2026.1899729

Tanzania house-design IAQ 2026 Research Square*A comparative study to investigate the impact of novel house design on indoor air quality in rural Tanzania*

PREPRINT. Magoda village, four novel-design houses against three local control houses over 77 days. Mean indoor PM_{2.5} was 2.8x lower and PM₁₀ 2.4x lower in the novel houses; the authors attribute this to integrated chimneys, increased envelope permeability, the lighting strategy and raised bedrooms, in a setting where ambient PM is low so indoor sources dominate. $n = 7$ houses is the limitation and it is a hard one: households are not randomised, occupancy and cooking behaviour differ, and the design elements are confounded with each other so no single feature is attributable. The secondary proposal - an hourly rather than 24-hour indoor guideline for homes empty during the day - is the more transferable idea

doi: 10.21203/rs.3.rs-10498557/v1

Turkish cigarette PM chamber 2026 Tob Use Insights*Particulate Matter Emissions of Turkish Cigarettes in Comparison to German Brands - An Aerosol Spectrometric Study*

Eleven Turkish brands, four German comparators and the 3R4F reference burned in a 2.88 m³ chamber. PM₁₀ 1044–1677 $\mu\text{g m}^{-3}$ and PM_{2.5} 1043–1675 $\mu\text{g m}^{-3}$, i.e. essentially all mass below 2.5 μm , with PM₁ 994–1548 $\mu\text{g m}^{-3}$ carrying the bulk. Five Turkish brands exceeded the reference cigarette on PM₁₀ and PM_{2.5}; all four German brands were lower. No correlation with tobacco weight, filling density, filter length or rod length, so the brand differences are not explained by the physical parameters measured. Limitation: a sealed unventilated chamber sized like a car cabin is a worst case, not a room, and the authors use it to argue for vehicle smoking bans rather than generalising to dwellings

doi: 10.1177/1179173x261473110

Burden, policy & mitigation

14 records

CARRIED FROM THE DAILY ISSUES — FULL SUMMARIES (14 OF 14 RECORDS IN THIS SUBTOPIC)

Addis Ababa hybrid net 2026 Health Serv Insights*PM_{2.5} spatiotemporal variation and monetised burden from a hybrid sensor network*

Two years of hourly PM_{2.5} (2022–2023) from a **hybrid network of seven low-cost sensors and two beta attenuation monitors** — the design this project cares about, deployed in a city with almost no prior monitoring. Site means **15–33 $\mu\text{g}/\text{m}^3$** , citywide **27**, more than fivefold the WHO guideline of 5; concentrations were higher in the wet Kiremt season, with a **07:00 diurnal peak of 38**. Hazard quotients exceeded 1 in all age groups. The cancer-risk estimate is honestly reported as unstable: ELCR exceeded 10^{-4} under a slope factor of 0.008 but stayed acceptable at 0.0012. Monetised DALYs give **USD 652.95 M** over two years, **0.25% of GDP** in each year, dominated by ischaemic heart disease. The paper does not describe co-location or humidity correction for the seven low-cost nodes, and every health figure is transferred from external risk coefficients, so the burden estimate inherits both instrument bias and transfer error.

doi: 10.1177/11786329261487535

Australian air health risk index 2026 Environ Pollut*METADATA ONLY — community-specific weather and air health risk indices*

No abstract retrievable on the entry date, despite *Environ Pollut* being PubMed-indexed — the same Crossref-ahead-of-deposit lag seen elsewhere in this issue. The title commits to community- and outcome-specific weather-and-air health risk indices built with explainable machine learning across **66 Australian communities**. The pollutant set, the health outcomes, the index construction and its skill are all unrecoverable and none is asserted here. Outcome-specific and community-specific index construction is the methodological direction that would replace one-size AQI banding, and 66 communities is enough to test whether the community-specific part earns its complexity; flagged for retrieval.

doi: 10.1016/j.envpol.2026.129117

CA wine-grape smoke 2026 (preprint) arXiv (preprint)*Quantifying impacts of wildfire smoke on crop yields in California*

Statistical regression of wine-grape yield on particulate matter alongside climate predictors at three Northern California sites. PM entered as a significant predictor jointly with temperature at two of the three, with final models reaching $R^2 = 0.79$ at **Sonoma** and **0.66 at Napa** — read by the authors as a shift from climate-controlled to wildfire-and-climate-controlled yields. Three sites, yield as an annual aggregate, and no separation of smoke PM from background or from the heat and radiation anomalies that accompany fire weather; with that many candidate predictors, an R^2 of this size is weak evidence of a specific PM pathway. Included as the only record this window treating PM as an economic rather than a health exposure, and because the smoke-attribution problem it dodges is the same one source-resolved monitoring exists to solve. Not peer reviewed.

doi: 10.48550/arXiv.2609.04422

CLHLS exercise 2026 Front Public Health*Exercise, county industrial air pollution and all-cause mortality in older adults*

16,954 CLHLS participants aged 65+, 23 provinces, three waves over six years, Cox models with time-varying covariates and sequential adjustment. Habitual exercise **HR 0.835 (0.773–0.902)**; county industrial SO₂ **HR 1.001 (0.874–1.146)** after adjustment; soot indicators null; and **no significant interaction** between exercise and any of the four pollution indicators. The null is about the exposure metric as much as the hypothesis: annual county soot and SO₂ emission and removal tonnage is an administrative aggregate, not a concentration, and it carries neither PM_{2.5} nor any within-county gradient. Exercise is self-reported and reverse causation from subclinical illness is not addressed. Reported here as a bounded null, not as evidence against interaction.

doi: 10.3389/fpubh.2026.1894835

COVID air-quality lessons review 2026 Preprints.org*COVID-19 Pandemic as an Environmental Health Indicator: Lessons Learned from Air Pollution Changes and Climate Implications*

PREPRINT, single author. Reviews the lockdown natural experiment: short-term outdoor air-quality improvement and greenhouse-gas reduction across many regions, the reported interactions between meteorology, pollutant concentrations and respiratory viral transmission, and the argument that temporary activity reductions demonstrate the ceiling of behavioural mitigation rather than a strategy. No systematic search, no inclusion criteria, no quantitative synthesis, not peer reviewed, and the pollution-COVID-transmission literature it draws on is itself heavily confounded by the timing of interventions. Logged for completeness of the burden-and-policy subtopic; carries no estimate this archive should propagate

doi: 10.20944/preprints202609.0712.v1

China PM₁ burden projection 2026 Environ Pollut*Ambient PM₁-Related Health Burdens and Mortality Influencing Factors in China: Historical Trends to Future Projections*

METADATA ONLY. No abstract was available on the entry date from any of the four sources queried. Title establishes a China-wide PM₁ attributable-burden analysis running historical trends into projections with a decomposition of mortality drivers. No exposure product, no concentration-response function, no scenario definition and no burden figure are available, so nothing is asserted and the record is tier C. Worth re-reading: PM₁ burden work is scarce because almost no regulatory network measures PM₁, which makes whatever exposure surface this study used the interesting part

doi: 10.1016/j.envpol.2026.129144

China county ageing vs PM_{2.5} 2026 GeoHealth*Association of Environmental Factors and Aging Levels Under Different Urbanization Levels in China*

County-level Chinese data for 2010 and 2020, 22 candidate factors, fixed-effect and random-forest models against the 65+ and 85+ population shares, stratified by high, medium and low urbanisation. Ageing levels rose 1.5–2.1 fold over the decade. PM_{2.5} and resident savings rank among the important variables in low-urbanisation counties; in medium-urbanisation counties CO, SO₂, PM_{2.5} and O₃ all enter; in high-urbanisation counties January and July mean temperature dominate instead. The design cannot support a health claim: ageing share is a demographic composition driven chiefly by migration and fertility, and variable-importance rankings from random forests on two cross-sections are not effect estimates. Carried for the policy-facing exposure gradient, not for causation.

doi: 10.1029/2025gh001642

Flue-gas condensable PM removal 2026 J Hazard Mater*Removing the particulate mass that filterable-PM controls never see*

Condensable particulate matter leaves coal-fired and municipal-waste-incineration stacks as vapour and forms particles in the atmosphere downstream, which is why filterable-PM control devices and, in many jurisdictions, the compliance measurement itself miss it entirely. This bench study passes flue gas through a string-of-beads liquid and reports **29.56–65.27% overall CPM removal at 1.17–8.25% of the energy** of conventional condensation. The organic fraction is removed more readily than the inorganic throughout, with the sulfate fraction best at **78.34%**. Mechanistically, condensation dominates at high flue-gas temperature and dissolution takes over as the gas cools, and a two-term model combining them reproduces the measured removal. It is bench scale with synthetic conditions — no long-duration fouling, no liquid regeneration economics, no full-scale pressure drop — so the efficiency range is an upper bound on what a retrofit would deliver.

doi: 10.1016/j.jhazmat.2026.143505

Hohhot park PM removal 2026 Environ Pollut*Random-forest attribution of PM reduction by urban park green space*

Thirty-seven parks in Hohhot, summer daytime, reduction rates computed for **six PM size fractions**. Linear, GAM and random-forest models were compared under leave-one-park-out cross-validation and RF won, though the fit is modest and honestly reported — point-level R^2 **0.19–0.58**, park-level **0.18–0.72**. SHAP plus the derivatives of partial-dependence and ICE curves identified nine drivers and, more usefully, the **threshold ranges beyond which the marginal mitigation effect plateaus**: air temperature 22.08–33.26°C, RH 29.71–67.17%, wind speed 0.14–1.12 m/s, distance from centre 2.7–16.79 km, perimeter-area ratio 0.15–0.33, contiguity 0.24–0.48, aggregation index 78.46–86.85%. Summer daytime only, one semi-arid city, and the in-versus-out contrast cannot separate deposition from dispersion and shelter effects. The plateau structure is the transferable result; the specific thresholds are not.

doi: 10.1016/j.envpol.2026.129128

Intergenerational mortality equity 2026 Environ Int*METADATA ONLY — intergenerational equity in acute and chronic mortality impacts*

No abstract retrievable on the entry date from Crossref per-DOI, Europe PMC or PubMed. What the record establishes: mortality impacts of *both* acute and chronic air pollution exposure have been assessed under an intergenerational health-equity framing — an unusual decomposition, since acute and chronic burden estimates are normally produced by separate literatures with incompatible baselines. What it does not establish: the populations, the pollutant set, whether PM is resolved, the discounting or life-table assumptions that an intergenerational comparison entirely depends on, the geography, or any result. None is asserted here. Flagged for retrieval as the highest-value of the four metadata-only records in this issue.

doi: 10.1016/j.envint.2026.110515

Livestock ammonia review 2026 Preprints.org*Livestock Ammonia Emissions: Impacts on Air Quality, Animal and Human Health, and Mitigation Strategies*

PREPRINT, single author. Reviews NH_3 formation from urea and uric acid hydrolysis and the physical controls on volatilisation - pH, temperature, moisture, residence time, exposed area, airflow - and insists on the distinction between concentration and mass flux, which is the measurement point that matters for inventories. Mitigation is framed as layered barriers: precision protein nutrition, rapid excreta removal, litter and slurry stabilisation, ventilation with capture, covered storage, low-emission land application; warming is noted to erode mitigation performance. No systematic search strategy, no quantitative synthesis, not peer reviewed. Relevant as agricultural NH_3 is the precursor controlling ammonium nitrate $\text{PM}_{2.5}$ across much of Europe and the US Midwest

doi: 10.20944/preprints202609.0590.v1

Puyang $\text{PM}_{2.5}$ -O₃ decade 2026 Preprints.org*A decade of diverging $\text{PM}_{2.5}$ and ozone in a Chinese oilfield city*

Preprint, not peer reviewed. Regulatory monitoring records for Puyang, one of the “2+26” channel cities, 2015–2025. $\text{PM}_{2.5}$ **fell 79.6 to 39.3 $\mu\text{g m}^{-3}$ while MDA8 ozone rose 123.0 to 189.6** — the canonical Chinese divergence, here in a city whose oil and gas VOC inventory makes it an unusually clean test of it. A positive $\text{PM}_{2.5}$ -O₃ correlation appeared only in the compound-pollution corner (O₃ > 160, $\text{PM}_{2.5}$ < 35), with seasonal r of 0.38 winter, 0.37 summer, 0.26 autumn, 0.21 spring — all weak. Ozone production efficiency declined over the record, correlated with $\text{PM}_{2.5}$ at $R = 0.46$ –0.65, and stayed above 7 monthly, which the authors read as NO_x-limited. OPE inferred from ambient monitoring rather than measured, no VOC speciation reported, and a single city; the control recommendation rests on that inference.

doi: 10.20944/preprints202609.0349.v1

TS-AQHI two-city validation 2026 Front Public Health*Development and two-city validation of a Temperature-Sensitive Air Quality Health Index incorporating synergistic effects of air pollution and thermal stress*

Weighted quantile sum regression supplies pollutant weights to a GAM carrying temperature, humidity and two-day-lagged multiplicative interactions; validated against daily mortality in Ningbo and London, 2014–2017. For high-risk categories the temperature-sensitive index gave excess risk 22.9 (95% CI 21.1–24.7) vs 15.0 (13.4–16.7) for conventional AQHI in Ningbo, and 22.2 (20.7–23.8) vs 7.7 (6.3–9.0) in London. Limitation: index weights are estimated and validated on the same 1,461-day series in each city with no out-of-sample or out-of-period hold-out, so the margin over the benchmark is an in-sample fit advantage until someone tests it forward. Two cities on different continents is nonetheless a real generalisation test of the framework

doi: 10.3389/fpubh.2026.1806127

Türkiye stroke GBD 2000–2023 2026 Neuroepidemiology*Ambient PM is the leading environmental risk for stroke death in Türkiye*

GBD 2023 secondary analysis for Türkiye. Age-standardised stroke incidence fell from 141.47 to 103.05 per 100,000 between 2000 and 2023, deaths from 132.61 to 57.03, and DALYs from 2696.30 to 1073.51 — a genuine and large decline. In 2023, metabolic risks accounted for 38.24% of stroke deaths (high blood pressure alone 21.29%) and smoking 12.20% among behavioural risks; **ambient particulate matter led the environmental risks at 11.93%**, placing it within a percentage point of tobacco. Macroeconomic regressions gave mixed and partly counterintuitive results — universal health coverage service capacity correlated positively with death rates — which is a familiar ecological artefact of ascertainment improving with health-system capacity. The deeper limitation for this watch: GBD’s PM exposure surface is itself a model, so the 11.93% is an attribution built on modelled exposure, not an independent measurement.

doi: 10.1159/ned/adwag030

Other clinical endpoints**4 records**

CARRIED FROM THE DAILY ISSUES — FULL SUMMARIES (4 OF 4 RECORDS IN THIS SUBTOPIC)

12M student visual impairment 2026 Environ Int*Associations of long-term exposure to $\text{PM}_{2.5}$ and its chemical constituents with incident visual impairment risk: A cohort study of 12 million school students*

METADATA ONLY. No abstract was available on the entry date from Crossref per-DOI, Europe PMC, PubMed efetch or Semantic Scholar - the same same-day Elsevier deposit lag logged repeatedly since 2026-09-08. Title, journal and cohort scale establish that this is a constituent-resolved long-term $\text{PM}_{2.5}$ cohort with an incident ocular endpoint in a 12-million-student population, which would make it the largest PM-vision cohort in the archive. No effect estimate, no exposure model description and no confounding structure are available, so nothing is asserted here and the record is tier C. Flag for re-reading once the abstract deposits

doi: 10.1016/j.envint.2026.110513

Kadoorie PM_{2.5} fracture/BMD 2026 J Hazard Mater

Associations of long-term exposure to ambient air pollution with fracture risk and bone mineral density: A cohort study of 0.5 million Chinese adults

490,172 China Kadoorie Biobank participants, 15,614 incident fractures over a median 12.0 years, with calcaneal quantitative-ultrasound BMD in 32,545. PM_{2.5} gave HR 1.09 (95% CI 1.05–1.13) per 10 $\mu\text{g m}^{-3}$ and warm-season O₃ 1.08 (1.04–1.12); a one-unit pollutant principal-component increment gave 1.08 (1.03–1.13), with PM_{2.5} contributing 70% by quantile g-computation. BMD fell 0.26–0.31 SD per PC increment and mediated only 5.8–8.6% of the fracture association, so the dominant pathway is not bone density. Exposure is assigned from the recruitment clinic's community, not the residence, and is fixed in space - a large cohort cannot repair ecological exposure assignment, and the 1 km product has no local validation.

doi: 10.1016/j.jhazmat.2026.143551

Multimorbidity meta-analysis 2026 BMJ Public Health

Pooled risk of multimorbidity per 5 $\mu\text{g m}^{-3}$ of PM and NO₂

Systematic review to 1 April 2026 across four databases (PROSPERO CRD42024613048); 22 studies included, 21 meta-analysed with random effects. Pooled adjusted risk ratios per 5 $\mu\text{g m}^{-3}$: PM_{2.5} **1.15 (1.08–1.23)** from 4 studies at $I^2 = 93.0\%$; PM₁₀ **1.27 (1.11–1.47)** from 3 at $I^2 = 99.5\%$; PM₁ **1.07 (1.05–1.09)** from 2 at $I^2 = 0.0\%$; NO₂ **1.12 (1.00–1.26)** from 4 at $I^2 = 96.6\%$. Nitrogen oxides, black carbon, SO₂, ozone and CO showed nothing. The limitation is structural, not incidental: harmonising to a per-5 $\mu\text{g m}^{-3}$ increment discarded most of the 22 studies, leaving two to four per pollutant, and at $I^2 = 93$ –99.5% the PM_{2.5} and PM₁₀ pooled values are averages over non-exchangeable studies. **Zero eligible studies** on polypharmacy — half the stated aim — or on household air pollution, older adults, the Americas or Oceania.

doi: 10.1136/bmjph-2025-004782

Tianjin myopia 2026 Environ Pollut

Long-term PM_{2.5} components and childhood myopia

School-based study of **212,566** children in grades 4–6 in Tianjin, with three-year mean PM_{2.5}, its major components (SO₄²⁻, NO₃⁻, NH₄⁺, organic matter, black carbon) and ozone estimated by machine-learning exposure models and assigned to *school* locations. Restricted cubic splines and quartile-based modified Poisson models for single pollutants, quantile g-computation for the mixture. **Sulfate carried the signal**: top versus bottom quartile **PR 1.10 (1.07–1.13)**, against a weak non-monotonic ozone association **1.03 (1.00–1.05)** and a whole-mixture estimate of only **1.017 (1.007–1.027)** per quartile increase. Cross-sectional, so myopia onset timing is unknown and reverse pathways through indoor time are not excluded; exposure at school ignores home and commute, which for a three-year window is substantial misclassification. The component contrast is the reason to read it — a speciated signal that unspeciated mass does not reproduce is the standing argument against mass-only networks.

doi: 10.1016/j.envpol.2026.129120

Pooled corpus register

First author	Focus	Journal	Design	Metric	Region
Neuro / mental health					
4HAIE hair cortisol	Intensity-dependent associations of physical activity and cardiorespiratory fitness with hair cortisol concentrations in the context of lifetime air pollution exposure	<i>Stress</i>	Cross-sectional (cohort baseline)	Lifetime residential PM _{2.5} , PM ₁₀ and NO ₂ from historical models	Czech Republic
ABCD amygdala 2026	Air Pollution, Early Adversity, and Amygdala: Environmental Correlates of Psychopathology in Preadolescents	<i>bioRxiv (preprint)</i>	Prospective cohort	Annual residential PM _{2.5} total mass plus 15 components	United States
ENVIRONAGE eye-track	Residential early-life air pollution exposure and oculomotor markers of cognitive workload in children: a prospective birth cohort study	<i>EBioMedicine</i>	Prospective cohort	Modelled residential BC, NO ₂ and PM _{2.5} ; windowed, per IQR	Europe
Older-adult cognition review	Air pollution and cognitive outcomes in older adults: a systematic review	<i>Rev Environ Health</i>	Systematic review	Ambient PM _{2.5} and co-pollutants, long-term	Global
Passive smoking acute stress	Is passive smoking associated with acute stress response? An observational real-world study	<i>Front Public Health</i>	Cross-sectional survey	Self-reported secondhand smoke exposure, no measurement	China
Sex modification dementia meta	Does Reported Sex Modifies the Association between Ambient Air Pollution exposure and Alzheimer's Disease and Related Dementia? A Systematic Review and Meta-Analysis	<i>Environ Pollut</i>	Systematic review + random-effects meta-analysis	Long-term PM _{2.5} and 11 further pollutants, F:M ratio	Global
WTC responder exposure mixture	Urban exposures, frailty, and mental illness in World Trade Center Health Program responders	<i>Nat Commun</i>	Prospective cohort (mixtures)	Annual PM _{2.5} , temperature, Green View Index	United States
Cardiovascular & metabolic					
CHARLS coexposure	Association of coexposure to indoor solid fuels, outdoor air pollution, and heatwave with the risk of Cardiometabolic Multimorbidity among middle-aged and older Chinese adults	<i>Int Arch Occup Environ Health</i>	Prospective cohort	PM ₁ , PM _{2.5} and PM ₁₀ with solid-fuel use and heatwave, Qgcomp mixture	China
California PM _{2.5} diabetes mortality	Associations of PM _{2.5} exposure with diabetes-related mortality for California residents	<i>Front Public Health</i>	Case-control (registry)	1-year rolling mean PM _{2.5} at geocoded residence	United States

First author	Focus	Journal	Design	Metric	Region
China WBC homeostasis	Long-term exposure to ambient air pollution and circulating white blood cell homeostasis among adults in China: Time-window-specific associations and the mediating role of fasting blood glucose	<i>Environ Pollut</i>	Retrospective cohort	1-, 6-, 12- and 24-month mean PM _{2.5} , PM ₁₀ , NO ₂ , SO ₂ , CO, O ₃	China
Constituent multiomics cardiopulmonary	PM _{2.5} Constituents and Cardiopulmonary Dysfunctions in Older Adults: A Multiomics Approach Unveiling Molecular Pathways	<i>Environ Sci Technol</i>	Prospective cohort (mixtures) + omics mediation	Personal PM _{2.5} monitoring with constituent speciation	China
Hangzhou AAD PM components	Associations of Fine Particulate Matter Components With Acute Aortic Dissection: A Time-Stratified Case-Crossover Study	<i>J Am Heart Assoc</i>	Time-stratified case-crossover	Component-resolved PM _{2.5} : BC, OM, sulfate, nitrate	China
Sichuan Basin CAA	Temporal changes in short-term PM _{2.5} -cardiorespiratory mortality associations and implications for the health benefits of the clean air action in the Sichuan Basin, 2014-2022	<i>Int Arch Occup Environ Health</i>	Time-series	Ambient PM _{2.5} , county daily means; ER per 10 ug m ⁻³	China
UK PH metabolomic mediation	Long-term ambient air pollution exposure, metabolomic signatures, and incident pulmonary hypertension: A large prospective cohort study	<i>Ecotoxical Environ Saf</i>	Prospective cohort (mixtures) + omics mediation	PM _{2.5} , PM ₁₀ , NO ₂ , NO _x and a joint air pollution score	UK
Zhongshan AMI	Daily changes in fine particulate matter are associated with acute myocardial infarction hospitalizations: an individual-level time-stratified case-crossover study in Zhongshan, China	<i>Front Public Health</i>	Case-crossover	Day-to-day PM _{2.5} change, CHAP 1 km, residence-assigned	China
Reproductive & developmental					
ECHO prenatal wildfire PM _{2.5}	Prenatal exposure to wildfire fine particulate matter and fetal growth outcomes in the United States: Findings from the ECHO Cohort	<i>Environ Epidemiol</i>	Prospective birth cohort + DLNM	Daily wildfire-specific PM _{2.5} , census-tract resolution	United States
France SUDI case-crossover	Short-Term Effect of Air Pollutants on Sudden Unexpected Death in Infancy: A Case-Crossover Study Based on the French National Registry	<i>Paediatr Perinat Epidemiol</i>	Time-stratified case-crossover	Modelled residential daily PM ₁₀ , PM _{2.5} , NO ₂ , O ₃	France
Korean birth-cohort growth mixture	Prenatal bisphenols and ambient air pollutants are differentially associated with early childhood length and weight from birth to 36 months: a multi-domain mixture analysis in a Korean birth cohort	<i>Environ Pollut</i>	Metadata only (no abstract)	Prenatal ambient air pollutants in a multi-domain mixture	South Korea
Preeclampsia nationwide 2018-23	Association between air pollution and preeclampsia: a nationwide study from 2018 to 2023	<i>Eur J Intern Med</i>	Metadata only (no abstract)	Ambient air pollution, nationwide registry linkage	Not stated
Prenatal greenspace ASD window	Mid-to-Late Pregnancy: A Critical Window for the Protective Effect of Green Space Against Autism Spectrum Disorder	<i>Environ Pollut</i>	Matched case-control + DLNM	Monthly residential PM _{2.5} alongside NDVI, -10 to +3 months	China
SIP prenatal Bayley	The effect of in utero exposure to air pollution and subsequent cognitive and behavioral development in children aged 1-3 years old	<i>Child Neuropsychol</i>	Prospective birth cohort	Prenatal PM _{2.5} and NO ₂ from public monitoring data	United States
Respiratory & allergic					
Amazon PM _{2.5} x heat	Interaction Effects of PM _{2.5} and High Temperatures on the Respiratory Health of Vulnerable Age Groups in the Brazilian Amazon	<i>Research Square</i>	Ecological panel	Monthly PM _{2.5} from biomass burning, temperature interaction	Brazil
Bengaluru COPD	Prevalence and Risk Factors of Chronic Obstructive Pulmonary Disease (COPD) in an Urban Poor Locality of Bengaluru, South India - A Cross-sectional Study	<i>Indian J Community Med</i>	Cross-sectional (multistage)	Self-reported cooking fuel, biomass and indoor smoke; no measurement	India
COPD anaemia metals	Air Pollution-Associated Intracellular Metals, Immune Gene Expression, and Erythrocyte Indices in COPD with Anemia	<i>Int J Chron Obstruct Pulmon Dis</i>	Repeated-measures biomarker	Annual PM ₁₀ , PM _{2.5} , NO ₂ , NO _x by land-use regression	Not stated
Cardiorespiratory review	Breathing in harm: current insights into the effects of air pollution on cardiorespiratory health	<i>Med Gas Res</i>	Narrative review	Ambient air pollution, no single exposure metric	Global
Disaster respiratory review	Respiratory emergencies in disasters: a narrative review	<i>Thorac Res Pract</i>	Narrative review	No new measurement; wildfire PM, silica and ash dust	Global / multi-region
Korea 8.25M asthma cohort	Cumulative Vulnerability to Particulate Matter and Incident Asthma: Joint Effects of Urbanicity, Poverty, and Age in a Nationwide Cohort	<i>Allergy</i>	Nationwide cohort	1-km long-term PM _{2.5} and PM ₁₀ estimates	South Korea

First author	Focus	Journal	Design	Metric	Region
Korea paediatric severity IV	Air pollution drives concentrated clinical severity of respiratory illness in early childhood	<i>medRxiv</i>	Registry cohort, FE models	Monthly PM _{2.5} , thermal-inversion instrument	South Korea
Kuwait tobacco smoke morbidity	Exposure to tobacco smoke, concurrent short-term symptoms of respiratory, allergic, cardiovascular morbidities and health-services use among young adults in Kuwait	<i>BMC Health Serv Res</i>	Cross-sectional survey	Self-reported first-hand and second-hand tobacco smoke	Kuwait
MobiliSense ambulatory spirometry	Evaluation of the effects of air pollutants on lung function using ambulatory monitor data from the MobiliSense Project	<i>Research Square</i>	Personal exposure monitoring + panel spirometry	Continuous personal BC and PM _{2.5} with GPS, 15 min-6 h lags	France
PM _{2.5} lung cancer never-smokers review	Breathing carcinogens: PM _{2.5} air pollution and lung cancer in never-smokers, a narrative review	<i>J Thorac Dis</i>	Narrative review	Narrative synthesis, PM _{2.5} exposure literature 2000-May 2026	Global / multi-region
Shijiazhuang AECOPD	Seasonal patterns and short-term environmental associations with AECOPD hospitalizations in Shijiazhuang, China, 2020-2024	<i>COPD</i>	Time-series	Daily PM _{2.5} / PM ₁₀ city means; lags 0-7 and 7-day mean	China
Umea wood smoke chamber	Acute controlled wood smoke exposure induces sensory airway irritation without detectable changes in small airway mechanics in healthy adults: a randomized crossover study	<i>Environ Res</i>	Randomised crossover chamber	Chamber PM1 453 +/- 13 ug m-3, 2 h vs filtered air	Europe
Mechanistic toxicology					
6PPD-Q AC16 p53	Exploring the targets and mechanisms of 6PPD-Q-induced cardiotoxicity: a network toxicology and molecular docking study	<i>Ecotoxicol Environ Saf</i>	In silico + in vitro	No airborne metric; 6PPD-Q is a road-dust constituent	China
AS-IV vs PM vascular senescence	Replicative and Air Pollution-Induced Senescence: Telomere Dysfunction, Mitochondrial Stress, and Modulatory Effects of Astragaloside IV in Endothelial and Vascular Smooth Muscle Cells	<i>Cardiovasc Toxicol</i>	Toxicological (in vitro)	Certified PM ₁₀ -like urban fine dust, 50 and 100 ug/mL	Computational (no site)
Berberine PM _{2.5} breast cancer	Berberine shows potential in mitigating PM _{2.5} -induced breast cancer progression by inducing DNA damage and inhibiting error-prone DNA repair pathways	<i>Apoptosis</i>	In vitro (cell lines)	Water-soluble extract of ambient PM _{2.5} (Delhi)	India
Fetal macrophage FCER1G	Prenatal PM _{2.5} exposure drives epigenetic reprogramming of fetal macrophages linked to atopic dermatitis	<i>Nat Commun</i>	Birth cohort + multi-omics	First-trimester ambient PM _{2.5} , modelled residential	South Korea
IL23A / NF-kB lung tumour	IL23A acts as a pivotal mediator of PM _{2.5} -induced lung tumorigenesis through the NF-kB pathway activation	<i>Ecotoxicol Environ Saf</i>	Animal + in vitro	PM _{2.5} extract exposure, NSCLC lines + orthotopic mice	China
Nasal PM _{2.5} testicular pyroptosis	Nasal Exposure to Environmental PM _{2.5} Drives Testicular Pyroptosis and Tissue Damage: Evidence from Inhibitor Intervention and Multi-Source Data	<i>Res Sq (preprint)</i>	Animal model	Intranasal PM _{2.5} , 10 mg/kg, 4 weeks	Computational (no site)
Retinal CRM	Environmental particulate reference materials increase DCFDA fluorescence and reduce ATP-dependent viability in primary mouse retinal cells	<i>Biochem Biophys Res</i>	Experimental / toxicology	Two certified environmental particulate reference materials, dosed	Laboratory
Tire wear particle phytotoxicity	Tire wear particles impact a roadside wild plant species	<i>Curr Biol</i>	Controlled exposure (ecotox)	Tire wear particles, non-exhaust traffic source	Japan
Vancouver DE blood-lung concordance	Concordant transcriptomics analysis of the lungs and blood after exposure to diesel exhaust: a randomized, controlled human exposure study	<i>Environ Res</i>	Randomised crossover chamber	Controlled 2-h diesel exhaust vs filtered air	Canada
Occupational & indoor					
Auditorium IAQ + 16S bioaerosol	Indoor Air Quality (IAQ) analysis and 16S rRNA gene sequencing of indoor air pollutants in a 3000-capacity religious auditorium	<i>Sci Total Environ</i>	Metadata only (no abstract)	Metadata only (no abstract deposited)	Global / multi-region
Bi/Bi2Te3 workplace ICP-OES	Determination of bismuth and bismuth telluride in workplace air by ultra microwave digestion with inductively coupled plasma optical emission spectrometry	<i>Zhonghua Lao Dong Wei Sheng Zhi Ye Bing Za Zhi</i>	Analytical method validation	MCE filter, 75 L; Bi and Te by ICP-OES after digestion	China
CFRP inhalation risk review	Inhalation risks of carbon fiber-reinforced plastic-derived emissions: Occupational exposure and regulatory perspectives	<i>Regul Toxicol Pharmacol</i>	Critical review	Respirable CFRP fibre fragments and thermal ultrafines	Global

First author	Focus	Journal	Design	Metric	Region
China gym co-exposure	Association of indoor multiple air-borne pollutants co-exposure and typical respiratory symptoms among gym employees: A nationwide observational study in China	<i>Environ Pollut</i>	Cross-sectional survey	Indoor PM _{2.5} , PM ₁₀ , total fungal and total plate count	China
Dutch school air cleaner RCT	Evaluating air cleaner effectiveness in schools: a comprehensive review and protocol of a cluster-randomized controlled trial of physicochemical and microbial markers	<i>Front Public Health</i>	Cluster-randomised trial protocol	Continuous PM ₁₀ , PM ₄ , PM _{2.5} , PM ₁ plus CO ₂ , T, RH, VOC	Netherlands
Ethiopia dormitory ventilation	Occupant-driven ventilation and indoor exposure dynamics in naturally ventilated Ethiopian university dormitories	<i>Indoor Environments</i>	Metadata only (no abstract)	Indoor exposure under occupant-driven natural ventilation	Ethiopia
Europe indoor PAH children	Polycyclic aromatic hydrocarbon (PAH) levels in indoor air and their associated health risks for children across Europe	<i>Environ Res</i>	Multi-city indoor survey	Active indoor air sampling plus deposited dust, 12 PAHs	Europe
JECS indoor PM Kawasaki	Household-specific indoor PM _{2.5} and PM _{10-2.5} concentrations and Kawasaki disease onset by age six: the Japan Environment and Children's Study	<i>Int J Hyg Environ Health</i>	Birth-cohort sub-study with indoor measurement	Household indoor PM _{2.5} and PM _{10-2.5} , 1-week gravimetric at ages 1.5 and 3.0 y	Japan
Occupational nanoparticle scoping	Occupational nanoparticle exposure and its effects on pulmonary health, cardiovascular health, and systemic biomarkers: a scoping review	<i>Part Fibre Toxicol</i>	Scoping review	Engineered nanomaterials and workplace ultrafine particles	Global
Oklahoma cannabis facility IAQ	Multi-parameter characterization of indoor air quality in a medical marijuana cultivation facility in Oklahoma: a pilot study of occupational exposure determinants	<i>Ann Work Expo Health</i>	Multi-instrument field evaluation	Size-resolved PM by TSI AirAssure + 2 PurpleAir PA-II-FLEX	United States
PSH smoke-free home RCT	Smoke-Free Home Intervention in Permanent Supportive Housing: A Cluster Randomized Clinical Trial	<i>JAMA Netw Open</i>	Cluster-randomised controlled trial	Secondhand smoke in multi-unit housing, CO-verified	USA
Rouyn-Noranda citizen dust	Metal contamination in residential snow and indoor dust near the Horne Smelter: a citizen-science study in Rouyn-Noranda, Quebec	<i>Environ Monit Assess</i>	Citizen-science sampling	Cu, Pb, Cd, As, Zn, Cr in snow and settled indoor dust	Canada
Tangshan CWP IGF1	Effects of dust exposure on IGF1 and autophagy related to coal workers' pneumoconiosis	<i>Front Public Health</i>	Experimental / toxicology	Cumulative occupational coal dust by job history; SiO ₂ in vitro	China
Tanzania house-design IAQ	A comparative study to investigate the impact of novel house design on indoor air quality in rural Tanzania	<i>Research Square</i>	Comparative field monitoring (built environment)	Continuous indoor PM _{2.5} and PM ₁₀ , 77-day deployment	Tanzania
Turkish cigarette PM chamber	Particulate Matter Emissions of Turkish Cigarettes in Comparison to German Brands - An Aerosol Spectrometric Study	<i>Tob Use Insights</i>	Chamber emission characterisation	PM ₁₀ , PM _{2.5} , PM ₁ by laser aerosol spectroscopy, 2.88 m ³ chamber	Turkiye
Sensing, forecasting & instrumentation					
3D-LSS particle shape	Enhanced three-dimensional light scattering sensor (3D-LSS) for online characterization of aerosol particle shape: Development and validation with test aerosols	<i>J Aerosol Sci</i>	Metadata only (no abstract)	Angularly resolved elastic scattering, single particle	Not stated
Aerosol core/morphology optics	Coupled Effects of Core Material and Aggregate Morphology on the Optical Properties of Atmospheric Aerosols	<i>J Aerosol Sci</i>	Metadata only (no abstract)	Metadata only (no abstract deposited)	Computational (no site)
Alkylamine IC-MS/MS method	An accurate and sensitive method for identifying atmospheric alkylamines: selective enrichment by alkaline purging-acidic absorption coupled with ion chromatography-tandem mass spectrometry	<i>J Chromatogr A</i>	Analytical method validation	Speciated gas and PM-phase alkylamines by IC-MS/MS	China
BC coating optics	Effects of V ₂ O ₅ , NiO, and Fe ₂ O ₃ surface coatings on the optical properties of black carbon aggregates	<i>J Aerosol Sci</i>	Metadata only (no abstract)	Optical properties of coated BC aggregates	Laboratory
Baltic ship plume OFR aging	In Situ Collection and Photochemical Aging of Individual Ship Plumes Captured at Sea	<i>Environ Sci Technol Lett</i>	Field campaign / shipborne	Shipborne oxidation flow reactor on individual vessel plumes	Europe
Bogota BiLSTM Huber	BiLSTM with asymmetric Huber loss for 24 h PM _{2.5} forecasting and early exceedance detection at an industrial monitoring station in Bogota, Colombia	<i>Atmosphere</i>	Deep-learning forecasting	Hourly PM _{2.5} , Puente Aranda industrial corridor	Colombia

First author	Focus	Journal	Design	Metric	Region
Boston sparse network	Air quality in cities: what do sparse sensor networks miss?	<i>Environ Sci Technol</i>	Mobile vs fixed co-measurement	Fixed-station PM _{2.5} vs mobile within a 250 m radius	United States
CALIOP Shanghai aerosol stratification	Long-term CALIOP observations of vertical aerosol stratification over Shanghai: Meteorological modulation and height-resolved transport	<i>Atmos Res</i>	Metadata only (no abstract)	Metadata only (no abstract deposited); CALIOP space-borne lidar	China
Climate-stratified PM _{2.5} LightGBM	Climate-stratified assessment of PM _{2.5} using machine learning: Geographic controls dominate meteorological factors	<i>PLoS One</i>	Machine-learning model	MERRA-2 satellite-assimilated monthly PM _{2.5}	Global / multi-region
Cooking aerosol FI-GAERO partitioning	Unravelling gas-particle partitioning dynamics in cooking aerosol oxidation through FIGAERO-CIMS analysis	<i>Atmos Chem Phys</i>	Aerosol flow reactor + model implementation	FIGAERO-I-CIMS speciation, 2-D VBS volatility	China
Coriolis sampler inactivation	Sampler-induced loss of respiratory virus infectivity revealed by spiked collection-liquid experiments	<i>Indoor Air</i>	Laboratory sampler validation	Coriolis micro liquid impinger, spiked collection liquid	Laboratory
Cyprus dust typing + DRE	Characterization of Saharan and Middle Eastern dust aerosols and their associated shortwave direct radiative effects over Cyprus	<i>Atmos Meas Tech</i>	Field campaign / remote sensing	AERONET optical properties, LIDAR, MODIS, HYSPLIT typing	Cyprus
Europe BLH regression kriging	Modelling monthly Boundary Layer Height maps combining radiosonde, satellite, and reanalysis over Europe	<i>Atmos Meas Tech</i>	Regression kriging map product	Gridded monthly PBL height, 25 km, 00 and 12 UTC	Europe
FY-4B/AGRI AOD evaluation	Evaluation of the FY-4B/AGRI Land Aerosol Optical Depth Product and Comparison with Geostationary Satellite Products over Asia	<i>Atmos Environ</i>	Metadata only (no abstract)	Geostationary land AOD retrieval, FY-4B/AGRI vs peer products	Global
Genoa absorption intercomparison	Multi-Wavelength Optical Characterization of Carbonaceous Aerosols at a Coastal Urban Site in Genoa: Intercomparison of Real-Time and Filter-Based Techniques	<i>Atmosphere</i>	Multi-instrument field evaluation	AE33 aethalometer vs MWA and BLAnCA filter analysis	Italy
India winter PM _{2.5} seasonal forecast	Seasonal predictability of winter PM _{2.5} pollution severity in India through a strong pollution-extratropical storm connection	<i>Sci Adv</i>	Seasonal prediction framework	Government reference PM _{2.5} network, >10 winters	India
LCS field calibration correction	Correction: Long-term performance evaluation and seasonal field calibration of a low-cost particulate matter sensor	<i>J Environ Health Sci Eng</i>	Metadata only (no abstract)	Low-cost PM sensor, seasonal field calibration	Iran
NaCl aerosol turbulent size redistribution	Turbulence-Induced Size Redistribution of Submicron NaCl Aerosols in Ribbed Confined Flows Prior to Electrostatic Collection	<i>J Aerosol Sci</i>	Metadata only (no abstract)	Metadata only (no abstract deposited)	Computational (no site)
Open-pit mine dust stacking model	Multi-Point Prediction and Interpretable Analysis of Dust Concentration in Open-Pit Coal Mine During the Cold Season Using a Stacking Learning Framework	<i>Atmos Pollut Res</i>	Metadata only (no abstract)	Multi-point cold-season mine dust concentration	Not stated
PPDRTL leaf-image AQI	PPDRTL: A novel framework for predicting pollutant deposition on roadside tree leaves using linear regression	<i>PLoS One</i>	Image-feature regression proxy	AQI regressed on leaf-surface image texture features	India
RI-URBANS urban AQ mapping	Enhancing Urban Air Quality Mapping through Novel Measurement and Modelling approaches, and Citizen Science: Actionable Insights from the RI-URBANS Project	<i>Atmos Meas Tech</i>	Multi-platform observation synthesis	Low-cost sensor networks, mobile platforms, citizen data	Europe
SAQIN	Long-short-term decoupled residual learning for high-resolution spatio-temporal air quality inference	<i>Neural Netw</i>	Machine-learning model	Criteria-pollutant fields inferred from sparse reference stations	Not stated
SE India monsoon aerosol	Monsoon-Regulated Aerosol Variability and Radiative Effects over a Rural Receptor Site in Southeast India	<i>Atmosphere</i>	Long-term field measurement campaign	MICROTOPS-II sunphotometer AOD500 and Angstrom exponent	India
SIBS inexpensive single-particle	Feasibility of an inexpensive single-particle SIBS instrument	<i>Aerosol Sci Technol</i>	Metadata only (no abstract)	Single-particle spark-induced breakdown spectroscopy	Not stated

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Shenzhen tower vertical PM _{2.5} DL	Leveraging multi-source observations for vertical PM _{2.5} estimation: A deep learning fusion model	<i>J Hazard Mater</i>	Data-fusion / ML surrogate model	Lidar extinction, microwave radiometer profiles, tower PM _{2.5} at 200/300 m	China
TROPOMI SO ₂ EAKF assimilation	Assimilation of SO ₂ TROPOMI Retrievals at the European Scale with EAKF Implemented in MINNI Through DART	<i>Atmosphere</i>	Satellite + chemical transport model	TROPOMI SO ₂ -COBRA total column into MINNI CTM	Europe
UAV single-particle CCSEM	Temporal evolution and vertical distribution of size-resolved aerosol composition and mixing state within the boundary layer using an unmanned aerial vehicle	<i>Environ Sci Technol</i>	UAV instrument deployment	Microscopy sampler on a UAV, 30 m and 1 min resolution	Global / multi-region
US precipitation washout DoubleML	Particulate matter removal by precipitation under normal and wildfire conditions across the United States using double machine learning	<i>Research Square</i>	Causal ML on monitoring records	Hourly EPA PM _{2.5} and PM ₁₀ vs NLDAS-2 precipitation	USA
W Sub-Saharan Africa PM monitoring	Monitoring and estimating PM _{2.5} and PM ₁₀ concentrations in Western Sub-Saharan Africa: Challenges and perspectives	<i>Atmos Pollut Res</i>	Metadata only (no abstract)	Regional PM _{2.5} /PM ₁₀ monitoring and estimation capacity	West Africa
Wildfire CHON PM _{2.5}	Integrating FT-ICR MS with Rule-Based Prediction and QSAR to Resolve Nitrogen-Containing Organic Compounds and Their Toxicity in Wildfire-Derived PM _{2.5}	<i>Environ Res</i>	Measurement campaign	Wildfire PM _{2.5} by GC-MS and FT-ICR MS; nitrated PAC in ng m ⁻³	South Korea
cCPC/OPC/SPS30 field evaluation	Field Evaluation of Community Condensation Particle Counters, Optical Particle Counters, and a Low-Cost Particle Sensor	<i>Aerosol Sci Technol</i>	Multi-instrument field evaluation	cCPC 0.005-2.5 um number; OPC 0.3-10 um; Sensirion SPS30 mass	United States
Exposure assessment & modelling					
Accra traffic intersections	Meteorological drivers of ambient PM _{2.5} pollution at urban traffic intersections in Accra, Ghana	<i>Environ Monit Assess</i>	Portable sensor monitoring campaign	Aeroqual Ranger portable dust monitor, 9 traffic hotspots	Ghana
Benin PM _{2.5} -water grid screening	Grid-based screening of fine particulate matter and water-optical conditions in Southern Benin: the Cotonou, Lake Nokoue, and Porto-Novo Corridor	<i>Environ Monit Assess</i>	Satellite remote sensing	Satellite-derived annual PM _{2.5} , 1 km grid; Sentinel-2 water optics	Benin
Brake wear SO ₂ uptake	Unexpectedly rapid SO ₂ heterogeneous reaction on the surface of vehicular brake wear particles	<i>Nat Commun</i>	Heterogeneous kinetics experiment	Brake wear particles, SO ₂ uptake coefficient	China
Brisbane Aitken-mode PNC	Long-term changes in Aitken-mode particle number concentration in Brisbane: roles of meteorology, traffic, and biomass burning	<i>Environ Int</i>	Metadata only (no abstract)	Long-term Aitken-mode particle number concentration	Australia
Construction equipment PM-bound EPFR	Real-world emissions of PM-bound environmentally persistent free radicals and reactive radical species generation from non-road construction equipment	<i>Atmos Environ</i>	Metadata only (no abstract)	PM-bound environmentally persistent free radicals	Not stated
Douala households	What is the impact of the concentration of particulate matter on the respiratory health of households in urban Cameroon?	<i>Spat Spatiotemporal Epidemiol</i>	Observational exposure study	PM ₁₀ , PM ₅ , PM _{2.5} by OC 300 Dust Particle Laser, low-cost portable	Sub-Saharan Africa
EROD-HR/SOILHD Iberia	Impact of high-resolution soil erodibility datasets on dust simulations in WRF-Chem with the GOCART scheme	<i>Atmos Chem Phys</i>	Regional CTM evaluation	Modelled PM ₁₀ vs SINCLAIR network, 5 dust episodes	Spain
GPS+LCS schoolchild exposure model	Integrating GPS and Low-Cost Sensors to Enhance Personal PM _{2.5} Exposure Modeling for Schoolchildren	<i>Environ Pollut</i>	Personal monitoring + microenvironment model	Portable PM _{2.5} sensor with GPS, 918 person-days; BAM-1020 reference	Taiwan
Hong Kong GEMA PM _{2.5} mobility	Spatiotemporal associations between PM _{2.5} , greenspace, physical activity, and stress: a geographic ecological momentary assessment study in Hong Kong	<i>Environ Res</i>	Geographic ecological momentary assessment	Mobile personal PM _{2.5} , 5-600 min cumulative windows	China
Liupan tethered eBC	Measurement report: impacts of thermodynamic and dynamic processes on the vertical distribution of carbonaceous aerosols: lessons from in-situ observations at the eastern foothills of Liupan Mountains, Loess Plateau	<i>Atmos Chem Phys</i>	Tethered-balloon profiling	eBC and UVPM profiles, tethered balloon, July 23 and 24	China

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Milan pizza-oven dispersion	Assessment of Air Quality Impacts from Wood-Fired Pizza Ovens: Comparison with Major Urban Emission Sources	<i>Preprints.org</i>	CTM simulation + source attribution	Measured emission factors + dispersion modelling vs ARPA network	Italy
N China indoor Finf 48 homes	Multivariable Determinants of Indoor PM _{2.5} and Infiltration Factors in 48 Residential Buildings Across Eight Northern Chinese Cities: A Seasonal Monitoring and Linear Mixed-Effects Modelling Study	<i>Atmosphere</i>	Seasonal indoor-outdoor monitoring + LMM	Paired indoor/outdoor PM _{2.5} at 5 min, hourly aggregate	China
NZ tractor fleet PM inventory	A calibrated, uncertainty-aware, and spatially explicit emission inventory of New Zealand's agricultural diesel tractor fleet	<i>Sci Total Environ</i>	Emission inventory	Non-road machinery PM _{2.5} , Monte Carlo uncertainty	New Zealand
Paediatric exposome	Assessing environmental exposures in children: toward a comprehensive exposome approach	<i>Pediatr Res</i>	Narrative review	Survey of questionnaires, biomarkers, sensors and geospatial indicators	Global
Paris Saharan dust heatwave	Numerical study of a Saharan dust plume impact on radiation, urban climate and thermal comfort in the Paris region during an heatwave event	<i>Atmos Chem Phys</i>	Mesoscale simulation + campaign validation	CAMS dust aerosol in Meso-NH, AOD-validated	France
St Elias proglacial dust	Modelling mineral dust emissions from proglacial valleys of the St. Elias Mountains, Canada	<i>Atmos Chem Phys</i>	Sub-km CTM development	Modelled dust emission vs in situ, camera and Doppler	Canada
Taiwan LRT bioaerosol	Impacts of long-range transport on the concentration, composition, and persistence of airborne fungi in northern Taiwan	<i>Environ Res</i>	Event-based monitoring	Daily culturable/spore fungal counts under ADS and FP	Taiwan
Taiwan PM _{2.5} -bound metals 71 sites	PM _{2.5} -Bound Metals in a High-Tech Industrial Area of Northern Taiwan: Spatial Distribution, Sources, and Health Risks	<i>Res Sq (preprint)</i>	Source apportionment	71-site PM _{2.5} filters, ICP-MS for 30 elements, PMF	Taiwan
Vilnius grey-green PM _{2.5} SHAP	Unraveling associations between grey-green spatial trade-offs and PM _{2.5} concentrations using explainable machine learning	<i>J Environ Manage</i>	Machine-learning model	Gridded PM _{2.5} 2016-2024; GGSCI, NDVI, nighttime lights	Lithuania
Xiamen chlorine NO ₂ uptake	Chlorine enhances nocturnal heterogeneous uptake of NO ₂ in coastal atmosphere under sea-land breeze circulation	<i>Atmos Chem Phys</i>	Kinetic model + field validation	Nocturnal particulate nitrate and HONO under sea-land breeze	China
Burden, policy & mitigation					
Addis Ababa hybrid net	Spatiotemporal variation of PM _{2.5} concentration and health impacts in Addis Ababa, Ethiopia: trends and patterns	<i>Health Serv Insights</i>	Network monitoring + burden	Hourly PM _{2.5} , 7 low-cost sensors plus 2 BAM, 2022-23	Ethiopia
Australian air health risk index	Constructing Community- and Outcome-specific Weather and Air Health Risk Indices Using Explainable Machine Learning: A Multicentre Study in 66 Australian Communities	<i>Environ Pollut</i>	Metadata only (no abstract)	Community- and outcome-specific air health risk index	Australia
CA wine-grape smoke 2026	Quantifying Impacts of Wildfire Smoke on Crop Yields in California	<i>arXiv (preprint)</i>	Modelling / inventory	Regulatory PM as a yield predictor alongside temperature covariates	United States
CLHLS exercise	Exercise is associated with lower all-cause mortality among older adults: a national cohort survival analysis with county-level industrial air pollution indicators	<i>Front Public Health</i>	Prospective cohort	County annual industrial soot and SO ₂ emissions and removals	China
COVID air-quality lessons review	COVID-19 Pandemic as an Environmental Health Indicator: Lessons Learned from Air Pollution Changes and Climate Implications	<i>Preprints.org</i>	Narrative review	Lockdown-period ambient PM and co-pollutant changes	Global
China PM1 burden projection	Ambient PM1-Related Health Burdens and Mortality Influencing Factors in China: Historical Trends to Future Projections	<i>Environ Pollut</i>	Metadata only (no abstract)	Ambient PM1, historical reconstruction and projection	China
China county ageing vs PM _{2.5}	Association of Environmental Factors and Aging Levels Under Different Urbanization Levels in China	<i>GeoHealth</i>	Ecological panel	County-level annual PM _{2.5} , CO, SO ₂ , O ₃ ; 2010 and 2020	China
Flue-gas condensable PM removal	Study on the removal of condensable particulate matter from flue gas using string-of-beads liquid	<i>J Hazard Mater</i>	Bench / device experiment	Condensable PM from coal and MSW incineration flue gas	China

First author	Focus	Journal	Design	Metric	Region
Hohhot park PM removal	Quantifying the reduction in particulate matter by urban park green spaces using a random forest model: a case study of Hohhot, China	<i>Environ Pollut</i>	Cross-sectional + ML	Six PM size fractions, summer daytime, in/out of park	China
Intergenerational mortality equity	Assessing intergenerational health equity in the mortality impacts of acute and chronic air pollution exposure	<i>Environ Int</i>	Metadata only (no abstract)	Acute and chronic exposure, mortality impact attribution	Not stated
Livestock ammonia review	Livestock Ammonia Emissions: Impacts on Air Quality, Animal and Human Health, and Mitigation Strategies	<i>Preprints.org</i>	Narrative review	NH3 flux as secondary inorganic PM precursor	Global
Puyang PM _{2.5} -O ₃ decade	Long-Term Trends and Nonlinear Interactions of PM _{2.5} -O ₃ Compound Pollution in a Typical Oilfield City of the 2+26 Region	<i>Preprints.org</i>	Monitoring-record analysis	Regulatory PM _{2.5} and MDA8 O ₃ , Puyang, 2015-2025	China
TS-AQHI two-city validation	Development and two-city validation of a Temperature-Sensitive Air Quality Health Index incorporating synergistic effects of air pollution and thermal stress	<i>Front Public Health</i>	Metric development + retrospective application	Multi-pollutant WQS-weighted index with temperature interaction	Global
Turkiye stroke GBD 2000-2023	Trends and Determinants of Stroke Burden in Turkiye, 2000-2023: An Analysis Using Global Burden of Disease Data	<i>Neuroepidemiology</i>	GBD secondary analysis	GBD ambient particulate matter risk-attributable fraction	Turkiye
Other clinical endpoints					
12M student visual impairment	Associations of long-term exposure to PM _{2.5} and its chemical constituents with incident visual impairment risk: A cohort study of 12 million school students	<i>Environ Int</i>	Metadata only (no abstract)	Long-term PM _{2.5} and specified constituents	China
Kadoorie PM _{2.5} fracture/BMD	Associations of long-term exposure to ambient air pollution with fracture risk and bone mineral density: A cohort study of 0.5 million Chinese adults	<i>J Hazard Mater</i>	Prospective cohort (mixtures)	Annual PM _{2.5} , PM ₁₀ , NO ₂ , warm-season O ₃ at 1 km	China
Multimorbidity meta-analysis	Association between air pollution, multimorbidity and polypharmacy: a systematic review and meta-analysis	<i>BMJ Public Health</i>	Systematic review + random-effects meta-analysis	Long-term ambient PM _{2.5} /PM ₁₀ /PM ₁ /NO ₂ , per 5 µg m ⁻³	Global
Tianjin myopia	Association of long-term exposure to ambient air pollution and myopia in Chinese children	<i>Environ Pollut</i>	School-based cross-sectional	3-yr PM _{2.5} and components, ML model at school point	China

Provenance and method

Period **2026-09-06 to 2026-09-12**, seven daily issues pooled. **This rollup queried no literature API.** It aggregates `state/corpus/*.json`, `state/metrics.csv` and `state/trials.json` over `PMRW_START-PMRW_END`; every record was screened, summarised and DOI-gated at daily-issue time and is reproduced verbatim. The single exception is the trial watch, which was queried once for the whole window against ClinicalTrials.gov with a no-term control, because no daily window in this period had logged one. **126 records, 126 unique DOIs** — no duplicate publication this period. Daily totals: 15, 20, 18, 26, 16, 10, 21. **Week-over-week comparisons** are against the 30 August–5 September rollup (150 records), with 23–29 August (109), 16–22 August (112) and 9–15 August (134) used where a longer run is cited; the 26 July–1 August week is excluded from "record" claims because it contains the founding retrospective back-scan. **Effect estimates** are pooled as recorded, on heterogeneous contrasts, with no meta-analysis and no heterogeneity statistic; several are co-pollutant rather than PM and are labelled as such on the plot. **Figures** are regenerated over the pooled range, not composited from the daily images. **Labels** — subtopic, design, particle-metric, geography — are single-label by dominant contribution, assigned at daily-issue time, and are a judgement rather than a controlled vocabulary.